

Theme: ECO-BOTS

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Theme-01

Eco-Bots

By the end of the theme students will be able to:

- Identify the key components of an ecosystem, including biotic and abiotic factors.
- Explain the interactions and interdependence between the components of an ecosystem.
- Recognize the various types of pollution (e.g., plastic, chemical, noise) that can negatively impact ecosystems.
- Analyse the effects of different pollutants on the health and balance of ecosystems.
- Discuss the importance of protecting and preserving ecosystems to maintain biodiversity and ecological sustainability.
- Investigate the physical and mechanical properties of different materials, including density.
- Determine the suitability of various materials for constructing objects based on their properties.
- Explain the relationship between an object's density and its ability to float or sink in water.
- Understand the principles of buoyancy and how they relate to the design of floating objects.
- Analyse the factors that contribute to an object's stability and ability to float, such as weight and surface area.
- Apply the principles of buoyancy and weight vs. surface area to design and optimize a floating device.

UNIT 01 [Beginner]

Chapter 01: Mechanics

LESSON 01: Eco-Bots.

Time frame: 75 mins

Introduction story section

Begin the lesson by sharing the story. You can use various engaging strategies to grab students' attention, ignite their curiosity, and link the story to the lesson's goals.

1. Preview with a Hook

- **Strategy:** Begin with an intriguing question, a striking image, or a captivating excerpt from the story to spark curiosity.
- **Purpose:** This strategy aims to capture students' interest immediately, connecting them emotionally or intellectually to the story before it begins. It builds excitement and creates a desire to find out what happens next.
- **How to Implement:**
 - Start by identifying a compelling moment, question, or visual related to the story. For example, if the story involves a sea turtle in distress, ask, “What would you do if you found a sea turtle with a plastic bottle cap caught around its neck?”
 - Alternatively, display a powerful image (e.g., a photo of a turtle affected by plastic pollution) and invite students to think about how they would respond in this situation.
 - Lead a brief discussion based on their responses, then connect their ideas to the theme of the story. For example, explain, “Today’s story is about a sea turtle facing a difficult situation. Let’s find out what happens next.”

2. Make Predictions

- **Strategy:** Show the cover, title, and perhaps the first line of the story, and ask students to predict what they think the story will be about.
- **Purpose:** Making predictions activates students' prior knowledge and sets a purpose for reading, encouraging them to engage with the story actively to see if their predictions are correct.
- **How to Implement:**

- Begin by showing the cover of the book and reading the title out loud. Ask, “What do you think this story might be about based on the title and the cover image?”
- Share the first line of the story if it hints at the theme or conflict, and ask students to expand on their predictions.
- Encourage students to share different predictions and jot down some of their ideas on the board. This way, they’ll be curious to see which predictions are accurate as they read.

3. Set the Scene with a Visual

- **Strategy:** Show a picture or play a short audio clip related to the setting or theme of the story.
- **Purpose:** Visuals and sounds can help students build mental images, immersing them in the world of the story even before reading. This strategy creates anticipation and establishes a sense of place or mood for the story.
- **How to Implement:**
 - Select a visual (e.g., an image of a forest, beach, or bustling city street) or audio (e.g., ocean waves, forest sounds) that relates to the setting of the story.
 - Share the visual or play the audio for a few moments, then ask students, “What do you imagine this place is like? How would it feel to be there?”
 - After a brief discussion, connect this sensory experience to the story’s opening scene, setting the stage for students to enter the story’s world with vivid expectations.

Story Reflection

Purpose of Reflection: This reflection section encourages students to think critically about the story and its connection to the lesson objectives.

Questions Answers:

What was wrong with Shelly the turtle when Adam found her?

c) She was choking on a plastic cap.

What did Adam and his family do after helping Shelly?

c) They made it their mission to pick up as much plastic and litter as they could find.

How to Implement:

1. Think-Pair-Share

- **Example Question:** *What was wrong with Shelly the turtle when Adam found her?*
- **How to Implement:**

- Ask students to think individually about possible reasons why Shelly might be in trouble, based on the story context or what they know about threats to sea turtles.
 - Have students pair up and share their ideas. This gives them a chance to predict what might be wrong with Shelly before discussing the actual problem of plastic pollution.
 - Invite pairs to share their answers with the class and discuss the concept of pollution and how it affects marine animals.
-

2. Role-Playing Responses

- **Example Question:** *What did Adam and his family do after helping Shelly?*
 - **How to Implement:**
 - Have students take turns role-playing as Adam, his family members, or even Shelly the turtle. Ask them to act out what they might do after saving a turtle from pollution.
 - Encourage students to think about how they would feel after helping Shelly and whether they would take any actions to prevent similar situations in the future.
 - After role-playing, discuss the importance of small actions like picking up litter and how Adam’s family made it their mission to help protect the environment.
-

3. Text Evidence Challenge

- **Example Question:** *What was wrong with Shelly the turtle when Adam found her?*
 - **How to Implement:**
 - Challenge students to find specific lines or words in the story that describe Shelly’s problem. For example, they could underline or highlight the phrase “choking on a plastic cap” or similar descriptions.
 - Ask students to explain how this evidence shows the problem Shelly faced and why plastic pollution is dangerous for wildlife.
 - Discuss how using text evidence helps them understand the story’s message about environmental responsibility.
-

4. Open-Ended Questioning

- **Example Question:** *Why do you think Adam and his family decided to make picking up litter their mission?*
- **How to Implement:**
 - Ask students to think about why Adam and his family might feel strongly about picking up litter after saving Shelly. Encourage them to consider how personal experiences can motivate people to take action.

- Allow students to share different reasons they think Adam's family became more environmentally conscious.
- Emphasize the idea that witnessing a problem firsthand can inspire action, encouraging students to consider what they might do to help in similar situations.

Let's Think

1. **Question:** *Based on the image, it appears to depict a significant amount of marine waste and debris floating in the ocean. Think what might be causing this problem of marine waste in the sea?*



- **Possible Answers:**
 - People littering on beaches or in rivers, which carry trash into the ocean.
 - Ships dumping waste in the ocean.
 - Plastic products that don't decompose, like bottles and bags, ending up in the water.
 - Lack of recycling or proper waste disposal methods.
 -
- **Strategy: Brainstorm Web**
 - **How to Implement:** Begin by drawing a web diagram on the board with "Causes of Marine Waste" in the center. Ask students to brainstorm possible causes and add them as branches. Let them work in pairs to think of as many ideas as they can, then share with the class.
 - This visual organization will help students connect their thoughts and see the bigger picture of pollution sources.
 - **Engagement Tip:** Use fun, colored markers or stickers to make the web visually appealing. Consider adding an element of competition by seeing how many unique answers each group can come up with.

Question 2: Imagine Yourself Swimming in a Beautiful Coral Reef. What Would You Expect to See?

Possible Answers:

1. **Fish or marine animals swimming near plastic waste:** Students might envision how marine life interacts with pollution.
2. **Plastic bottles, bags, and other trash floating on the water's surface:** They could picture the direct impact of waste on the ocean's surface.
3. **Dirty water or polluted areas around the waste:** Students may consider how pollution affects water quality.
4. **Fewer plants or coral in areas with a lot of trash:** They might recognize the detrimental effects on marine ecosystems.

Strategy: Draw What You See

- **How to Implement:** Provide students with blank paper and colored pencils or markers. Instruct them to draw what they envision seeing underwater near a coral reef that is polluted with plastic waste. Encourage them to include marine life, trash, and the overall state of the water.
- **Engagement Tip:** Display the completed drawings around the classroom, creating a "pollution gallery." This will not only showcase their creativity but also serve as a visual reminder of the effects of pollution on marine life. After displaying their work, engage the class in a discussion about how pollution impacts marine ecosystems, using the students' artwork as a conversation starter.

3. Question: *In your opinion, how can we make less waste?*

- **Possible Answers:**
 - Use reusable bags, bottles, and containers instead of single-use plastic.
 - Recycle more and encourage others to do so.
 - Organize or participate in beach clean-up events.
 - Educate others about the importance of reducing waste.
- **Strategy: Action Plan Posters**
 - **How to Implement:** Ask students to work in small groups to create posters with practical steps for reducing waste. Each group can focus on one main idea (like recycling or avoiding plastic), then illustrate and write simple actions everyone can follow.
 - **Engagement Tip:** Offer fun materials like stickers, glitter, or stamps to decorate the posters. Hold a mini "poster fair" where groups present their ideas to the class and vote on the best waste-reduction strategy.

Explore

The concept of STEAM scouts is built around exploration, trial, and error. This section is divided into three parts: a "Do It" activity where students can experiment with the new concept firsthand, followed by a discovery phase where they uncover the scientific principles behind it.

Do it

Sorting Materials:

Objective: Students will explore and understand the properties of different materials (organic, plastic, and metal) through hands-on sorting activities, fostering their observation, classification, and critical thinking skills.

Activity Overview

Materials Needed:

- Three labeled pots or bins (Organic, Plastic, Metal)
 - A list of materials for sorting, which may include:
 - Food scraps (e.g., fruit peel, vegetable scraps)
 - Plastic items (e.g., plastic bottle, plastic fork, plastic wrap)
 - Metal items (e.g., aluminum can, metal spoon, paperclip)
 - Worksheet for recording observations (optional)
 - Markers or labels for identifying pots
 - A timer (optional)
-

Step-by-Step Activity Instructions

1. **Preparation:**
 - Set up three pots or bins labelled **Organic**, **Plastic**, and **Metal**. Ensure they are clearly visible to all students.
 - Gather various materials to sort, ensuring a good mix for each category. Place these materials on a table or a large surface accessible to students.
2. **Introduction:**
 - Explain the purpose of the activity to students. Discuss what they will be doing: sorting different materials into the correct pots based on their properties.
 - Introduce the concept of recycling and the importance of knowing the difference between types of materials.
3. **"Do It" Activity :**
 - **Sorting Instructions:**
 - Distribute the list of materials to each student or group.
 - Instruct students to pick up items from the table and decide which pot they belong to.
 - As they sort, encourage them to discuss with peers why they think an item fits into a specific category.

- Remind them that they can experiment and try different sorting methods, such as grouping similar items together.
 - 4. **Discovery Phase:**
 - Once sorting is complete, review each pot as a class.
 - Ask students to share their reasoning behind their sorting choices.
 - Discuss the properties of each material and its classification:
 - **Organic:** Biodegradable, comes from living organisms.
 - **Plastic:** Synthetic, made from petroleum-based products, often non-biodegradable.
 - **Metal:** Conductive, typically recyclable.
 - Highlight any discrepancies in sorting and encourage a discussion on why certain items may have been misclassified.
 - 5. **Reflection :**
 - Have students write down one new thing they learned about materials during the sorting activity.
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Teaching Strategy

- **Inquiry-Based Learning:**
 - This activity encourages students to ask questions and explore the properties of materials through hands-on experience.
 - By allowing students to engage in discussion, they can develop critical thinking and reasoning skills.
- **Collaboration:**
 - Encourage students to work in pairs or small groups. Collaboration fosters communication skills and helps students learn from each other.
- **Active Learning:**
 - Engaging students in physical sorting and discussion promotes active participation, which enhances retention and understanding of scientific concepts.
- **Formative Assessment:**
 - The reflections and group discussions serve as informal assessments of student understanding, allowing teachers to gauge learning outcomes and clarify concepts as needed.

Watch it

<https://www.youtube.com/watch?v=MnCbTTTi7ic>

- **Before Watching – Set a Purpose.**
 - **Preview Questions:** Introduce a few questions related to the video content, like "What types of materials do you see in the video?" or "How do people sort these materials in real life?" This primes students to watch actively for answers.
 - **Vocabulary Preview:** Go over any key terms students might encounter (e.g., "biodegradable," "recycling," "landfill"). This helps them understand the video better and builds anticipation for learning.

- **During the Video – Active Engagement Techniques.**

- **Watch & Pause (Chunking):** Divide the video into short segments by pausing after key points. Ask questions like, “What did you learn about plastic materials here?” or “Why is sorting materials important?” This keeps students thinking and discussing, rather than passively watching.
- **Note-Taking Worksheet:** Give students a simple worksheet with sections for each type of material (organic, plastic, metal). As they watch, they can jot down examples of each material type mentioned in the video.
- **Whisper Partners:** Occasionally, pause and have students discuss a question briefly with their neighbor. For example, after a segment on plastic waste, you could ask, “How is plastic affecting the environment? Turn to your partner and discuss.” This keeps them actively processing information.
- **Hand Signals for Key Points:** Tell students to give a silent hand signal (like a thumbs up) whenever they see something important or interesting. This keeps them attentive and provides an opportunity to review what caught their attention.

- **Post-Viewing Activities – Discussion and Reflection.**

- **Quick Group Discussion:** Immediately after the video, hold a brief class discussion. Ask, “What was something new you learned?” or “Which material do you think is the most harmful to the environment?” This provides an outlet for their ideas and interpretations.
- **Exit Ticket Reflection:** Have students write a quick reflection or "exit ticket" describing one action they can take to reduce waste at home, based on what they learned.
- **Creative Follow-Up (Optional):** Have students illustrate or design a poster on one of the materials and ways to manage its waste effectively. This reinforces concepts and lets them creatively express their understanding.

Read about it

<https://kids.nationalgeographic.com/nature/kids-vs-plastic/article/pollution-1>

Strategies for an Engaging "Read" Activity:

- **Pre-Reading Discussion.**

- **Introduction and Background Knowledge:** Start with a quick discussion about pollution and plastic waste to activate prior knowledge. Ask, “What do you already know about pollution and its effects?” or “Where have you seen plastic pollution in your daily life?” This primes them to connect with the article’s topic.
- **Prediction Activity:** Before reading, show students a few images from the article and ask them to predict what problems with plastic pollution might be discussed. This creates curiosity and sets a purpose for reading.

- **During Reading – Active Engagement Techniques**

- **"Stop and Jot" Notes:** Have students pause after each section to quickly jot down a one-sentence summary or a "wow" fact. For example, after reading about the impact of plastic on animals, they might note, "Animals mistake plastic for food, which can make them sick."
 - **Questioning Strategy:** Encourage students to write one question in the margin of each section. Questions could be, "Why is plastic so harmful?" or "What can I do to reduce plastic at home?" These questions can be shared in a group discussion afterward.
 - **Buddy Reading:** Pair students and assign each partner a paragraph or section to read aloud. Partners can then take turns asking each other questions about what they read or sharing one interesting fact they found in their section.
- **Post-Reading Activities – Reflection and Discussion**
- **Summarize and Share:** Ask students to summarize the article by writing down the main problem (plastic pollution) and one major solution or action they can take. This helps them consolidate their understanding.
 - **Think-Pair-Share:** Have students think of one key fact or surprising detail from the article, then pair up and discuss with a partner before sharing with the whole class. This builds confidence and reinforces information.
 - **Exit Ticket Reflection:** On a sticky note or index card, have students write one personal action they can take to reduce plastic use, based on the article. Collect these and post them as a class "Pledge Wall" to reinforce commitment to small, impactful actions.

Assessment

Focus





Practice

<https://www.youtube.com/watch?v=0iRTszLhD5M>.

Activity: Building a Paper Pyramid for the Marine Ecosystem

Objective: Students will construct a 3D pyramid to represent elements of the marine ecosystem, helping them visually organize and better understand different components of this ecosystem.

Materials Needed:

- Cardboard
- Glue gun and glue sticks
- Printed paper with images or facts about the marine ecosystem
- Scissors
- Ruler
- Pencil

Steps

- Cutting the Triangles:**
 - Using the cardboard, measure and cut out 8 triangles, each with dimensions of 19 cm (height) by 7.5 cm (base).
- Cutting the Square Base:**
 - Measure and cut a square with dimensions of 17.5 cm by 17.5 cm. This will serve as the base of the pyramid.
- Creating the Pyramid Sides:**
 - Take two of the triangles and glue them together along their edges so that they face each other. Repeat this for the other triangles, creating four pairs in total.
- Assembling the Pyramid:**
 - Glue the pairs of triangles together to form the sides of the pyramid. Once assembled, glue the entire structure onto the square base.
- Adding Printed Paper:**

- Place images or information about different marine ecosystem elements on each face of the pyramid, such as fish, coral reefs, marine plants, and water pollution effects.

Teacher's Strategies

- **Hands-On Learning:** This activity is hands-on, engaging students in a tactile project that reinforces their understanding of the marine ecosystem. By building the pyramid, students actively participate and are more likely to retain information.
- **Visual Aids for Reinforcement:** Using printed images or facts about marine life on each face of the pyramid provides visual aids, which help students connect the structural sides of the pyramid to specific parts of the marine ecosystem.
- **Collaboration and Peer Support:** Students can work in small groups, supporting each other with measuring, cutting, and assembling. This promotes teamwork, as well as communication and problem-solving skills.
- **Reflection and Discussion:** After constructing the pyramid, students can discuss what each side represents and explain why each part of the ecosystem is important. This reflection helps reinforce learning and allows students to share insights with their peers.
- **Real-World Connections:** By connecting the physical pyramid with elements of the marine ecosystem, students see how different parts of an ecosystem work together, encouraging them to think about the relationships within ecosystems and the importance of protecting marine life.

Challenge

This activity is an optional extension, intended to be introduced to students as a bonus task they can work on independently if they wish. It is not required to be completed during the session.

UNIT 01 [Beginner]

Chapter 01: Mechanics

LESSON 02: Engineering Design Process.

Time frame: 15mins

Story

Apply the same strategies used in Part 1 of the story.

Story Reflection

Understanding the Role of Environmental Engineers

Overview:

Environmental engineers help to protect and improve the environment by applying scientific principles to solve ecological challenges. Their work is essential in maintaining clean air, water, and soil, which directly impacts human health and ecosystems.

Objectives:

- Help students understand the importance of environmental engineering in safeguarding natural resources.
- Enable students to identify key projects and methods used by environmental engineers.
- Inspire students to think about careers in environmental science and engineering.

<https://www.youtube.com/watch?v=T2xFDIdjX88>

Engineering Design Process

Quickly review the phases of the Engineering Design Process, then encourage students to focus on the **Ask** and **Brainstorming** stages until they settle on a project idea. Remind them that there are no right or wrong answers for this step—it's all about exploring possibilities!

Ask

What solutions can we explore to address pollution in marine ecosystems?



Imagine

Building a water floater to collect waste.

UNIT 01 [Beginner]

Chapter 01: Mechanics

LESSON 03: Design Your Floater.

Time frame: 45 mins

Stepping Stone.

You need to know ..

Basic Principles of Engineering Design process.

Types of Materials.

Definition of Force.

You will be able to..

Build Waste floater.

Choose the best material and design for floater.

Define force of bouncy and see its effect on floater.

Story

Apply the same strategies used in Part 1 of the story.

Let's Think

In the **Think** section of the teacher guide, students will brainstorm ideas to answer key questions about designing a waste-collecting device for water. Here are possible responses and teaching strategies to guide this brainstorming session.

Questions and Possible Answers

1. **How does our floater stay above water?**

○ Possible Answers:

- Use lightweight, buoyant materials like plastic or foam.
- Design with a wide base to improve stability.
- Add air-filled compartments or sealed containers to increase buoyancy.

2. **What is our plan for collecting the waste?**

○ Possible Answers:

- Use a net or mesh to scoop waste from the water.
- Create a funnel system to direct floating waste into a collection area.
- Design a conveyor belt or rotating mechanism to gather waste efficiently.

3. **What method will we use to make it move in the water? Will it be manual or electric?**

○ Possible Answers:

- Attach paddles or small propellers powered by solar energy or batteries.
- Design a hand-operated paddle or lever system.

- Consider attaching it to a remote-controlled motor for precision.
 - 4. **How will we store the waste we've collected so it can be recycled?**
 - Possible Answers:
 - Use detachable compartments that can be emptied easily.
 - Design a container with sections to separate recyclable and non-recyclable items.
 - Include a cover or lid to prevent waste from falling back into the water.
-

Teaching Strategies for This Section

1. **Guided Brainstorming**
 - **Activity:** Lead students through each question, encouraging them to share their ideas. Write ideas on the board to visually map out possibilities.
 - **Purpose:** Helps students feel comfortable sharing different ideas, recognizing there is no single "right" answer, and fostering creativity.
2. **Think-Pair-Share**
 - **Activity:** Have students think individually, then discuss ideas with a partner before sharing with the class. For each question, ask a few pairs to share their ideas.
 - **Purpose:** Encourages collaborative thinking and allows quieter students to feel more confident before presenting to the group.
3. **Visual Brainstorming (Sketching)**
 - **Activity:** Ask students to sketch possible designs for each question, especially for flotation or movement ideas. Students can then share their drawings with the class.
 - **Purpose:** Visual thinking helps students conceptualize and communicate ideas they might struggle to explain verbally, enriching the brainstorming process.
4. **Question Prompts and Scaffolding**
 - **Activity:** For each question, provide additional prompts to guide deeper thinking (e.g., "What materials would float best?" or "How can we power this sustainably?").
 - **Purpose:** Supports critical thinking by encouraging students to consider factors like materials, power sources, and sustainability.
5. **Role-Based Ideation**
 - **Activity:** Assign each student or group a "role" (e.g., designer, engineer, environmentalist) and have them think from that perspective as they brainstorm.
 - **Purpose:** Fosters empathy and broadens students' viewpoints by helping them consider multiple angles in problem-solving.

Explore

Do it

<https://copilot.microsoft.com/>

Assist students in crafting an effective prompt that aligns with the specific eco-bot they envision.

Watch it

<https://www.youtube.com/watch?v=6j2IWCx7QR8>

To help students engage with this video effectively, you can use these strategies:

1. Visual Comparison Activity

- **Activity:** Have students watch the video and observe the real floater's design. After watching, prompt them to compare it with the eco-bot they imagined earlier.
- **Instructions:** Ask students to note similarities and differences between their designs and the real floater, focusing on structure, movement, and waste-collection methods.
- **Purpose:** This helps students refine their ideas, see practical engineering applications, and understand how their designs might be improved.

2. Guided Note-Taking

- **Activity:** Before starting the video, provide students with a simple worksheet or list of questions to guide their observation. Include prompts like:
 - What are the main parts of the floater?
 - How does it move in the water?
 - What materials are used?
 - How does it collect and store waste?
- **Purpose:** This helps students focus on specific elements while watching, reinforcing observation and attention to detail.

3. Group Discussion and Reflection

- **Activity:** After watching the video, organize students into small groups to discuss their observations. Encourage them to share how the real floater's parts and mechanisms compare with their ideas.
- **Purpose:** This promotes critical thinking and allows students to reflect on design feasibility, encouraging collaborative evaluation of ideas.

4. Parts Identification Exercise

- **Activity:** Ask students to write down the main parts of the floater seen in the video and briefly describe each part's purpose or function.
- **Purpose:** This deepens students' understanding of the floater's components, preparing them to integrate effective parts into their designs.

Read about it

Here are several reading strategies to use while students engage with the article about the AI-enabled robotic boat:

1. Previewing the Text

- **Activity:** Before reading, have students skim the article to get an overview. They can look at headings, subheadings, images, and captions.
- **Purpose:** This prepares students for the content and context, helping them formulate questions or predictions about what they will learn.

2. Chunking the Text

- **Activity:** Divide the article into sections (paragraphs or key ideas) and have students read each chunk separately, summarizing the main idea of each part before moving on.
- **Purpose:** This breaks down the reading into manageable parts, making it easier for students to process information and retain key concepts.

3. Think-Alouds

- **Activity:** Model think-alouds by verbalizing your thoughts as you read certain passages. Encourage students to do the same in pairs or small groups, sharing their thoughts on what they are reading.
- **Purpose:** This strategy helps students develop comprehension skills by demonstrating how to analyze and reflect on the text as they read.

4. Graphic Organizers

- **Activity:** Provide students with a graphic organizer, such as a Venn diagram or a concept map, to visually map out information as they read.
- **Purpose:** Visual aids help students organize their thoughts, clarify relationships between concepts, and synthesize information from the text.

5. Questioning the Text

- **Activity:** Have students generate questions about the content as they read. They can focus on the "who, what, when, where, why, and how" of the article.
- **Purpose:** This encourages critical thinking and helps students engage more deeply with the material, leading to better comprehension and retention.

6. Collaborative Reading

- **Activity:** Arrange students into small groups to read the article together, discussing and clarifying sections as they go.
- **Purpose:** Collaborative reading allows students to share insights and explanations, which can enhance understanding and build confidence.

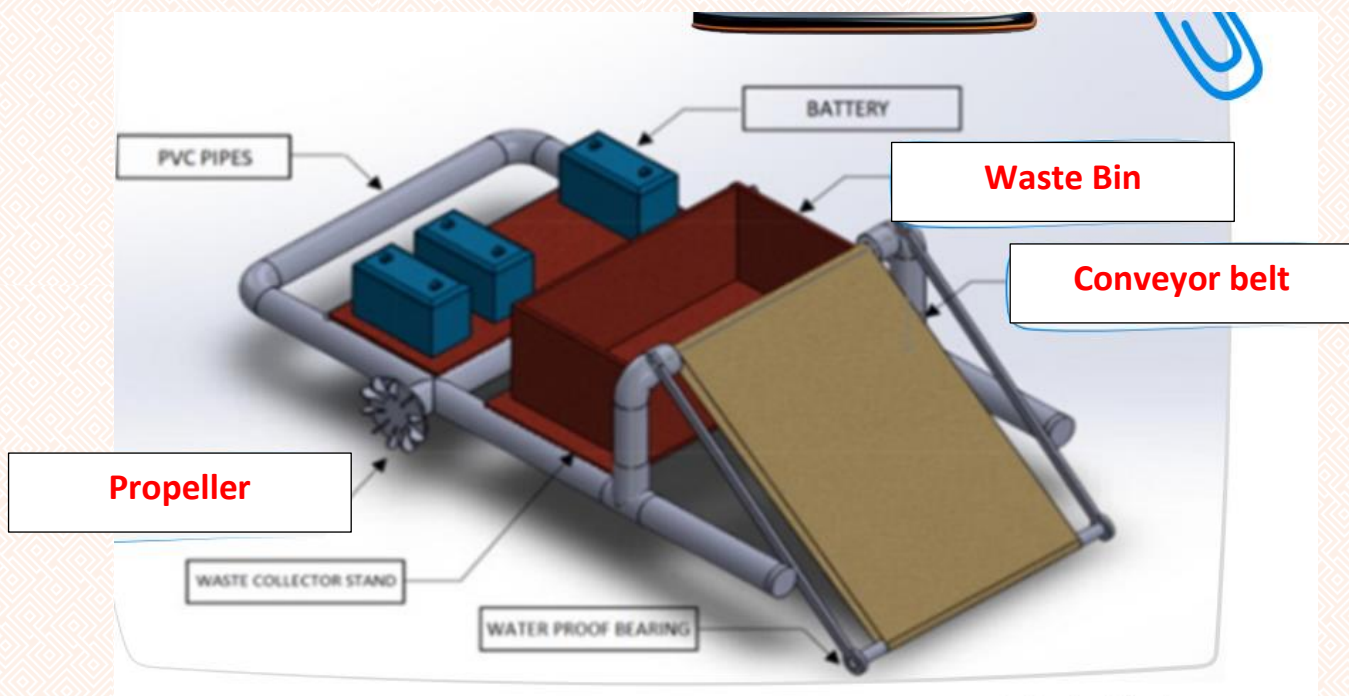
7. Post-Reading Reflection

- **Activity:** After reading, have students write a brief reflection on what they learned, including any surprising facts or ideas they found particularly interesting.
- **Purpose:** This encourages students to process the information and articulate their understanding, reinforcing key concepts from the article.

Assessment

Focus

Fill in the gaps to complete the floater parts.



For the question "Fill in the gaps to complete the floater parts," here are some effective strategies suitable for Grade 5 students:

1. Guided Group Activity

- **Activity:** Organize students into small groups and provide each group with a worksheet that includes sentences or diagrams with missing parts related to the floater's components. Encourage them to discuss and collaborate to fill in the gaps.
- **Purpose:** This promotes teamwork and communication skills while allowing students to learn from one another.

2. Visual Aids

- **Activity:** Use images or diagrams of a floater, highlighting different parts. As students fill in the gaps, they can refer to these visuals to understand how each part functions.
- **Purpose:** Visual aids help reinforce learning and make abstract concepts more concrete.

3. Think-Pair-Share

- **Activity:** Have students first think individually about what goes in the gaps. Then, they can pair up to discuss their thoughts before sharing with the class.
- **Purpose:** This encourages independent thinking followed by collaboration, allowing students to refine their ideas before presenting them.

4. Word Bank

- **Activity:** Provide a word bank with key terms related to the floater's parts. Students can use this bank to help fill in the gaps.
- **Purpose:** A word bank offers support, making it easier for students to recall terminology and apply it correctly.

Practice

Encourage students to use their imagination with this question.

Challenge

This activity is an optional extension, intended to be introduced to students as a bonus task they can work on independently if they wish. It is not required to be completed during the session.

EDP

Return back to the EDP paper and add the design they made and the parts they will add to the plan part.



UNIT 01 [Beginner]

Chapter 01: Mechanics

LESSON 04: Choose Your Materials.

Time frame: 45 mins

Story

Apply the same strategies used in Part 1 of the story.

Think and Do it

Sink or Float Activity

Objective:

Students will predict and test whether a variety of materials will sink or float in water.

Materials Needed:

- A container filled with water (e.g., a large tub or sink)
- The following items:
 - Plastic bottles
 - Metal nails
 - Wooden sticks
 - Rubber bands
 - Foam pieces
- Paper and pencils for recording observations

Steps:

1. **Introduction to Concepts:**
 - Begin by discussing the concepts of density with the students. Explain that objects denser than water will sink, while those less dense will float.
2. **List of Materials:**
 - Present the following list of materials to the students:
 - Plastic bottles
 - Metal nails
 - Wooden sticks
 - Rubber bands
 - Foam pieces
3. **Make Predictions:**
 - Ask students to predict whether each item will sink or float. Have them write their predictions in a table with two columns: **Sink** and **Float**. For example:

4. **Testing the Materials:**

- Instruct students to carefully place each item into the water one at a time.
- Observe the behavior of each item as they place it in the water.

5. **Record Observations:**

- Have students record their observations in their tables, marking whether each item sank or floated. For example:

Item	Sink Float
Plastic bottles	✓ ☐
Metal nails	✓ ☐
Wooden sticks	✓ ☐
Rubber bands	✓ ☐
Foam pieces	✓ ☐

6. **Discussion:**

- After testing all materials, hold a class discussion about the results. Ask questions like:
 - Which items floated? Why do you think that is?
 - Which items sank? What does that tell us about their density?
 - Were there any surprises in your predictions vs. the results?

7. **Conclusion:**

- Summarize the activity by reinforcing the concept and density. Discuss how understanding this concept can help in real-world applications, such as designing boats.

Explore

Watch it

<https://www.youtube.com/watch?v=2dyCe1GPagE>

To effectively explain the video and engage students, consider these strategies:

1. **Pre-Viewing Discussion:**

- Start with a brief discussion about the video topic. Ask students what they know about it or their initial thoughts. This activates prior knowledge and builds interest.

2. **Set a Purpose:**

- Give students specific questions to think about while watching the video. For example, “What are the main points discussed?” or “How does the video relate to our current topic?” This helps focus their attention.

3. **Pause for Reflection:**

- Pause the video at key moments to discuss what has been shown. Encourage students to share their thoughts and ask questions. This keeps them engaged and allows for deeper understanding.
- 4. **Interactive Elements:**
 - Incorporate interactive elements, such as polls or quizzes, during or after the video. Tools like Kahoot or Mentimeter can make this fun and engaging.

Do it

Which is the best floater

Materials Needed:

- Wood
- Foam
- Plastic Bottle
- PVC

Steps to Follow:

1. **Gather Materials:**
 - Obtain a transparent water basin or a large clear container.
 - Collect the materials you will be testing: wood, foam, plastic bottle, and PVC.
2. **Testing Wood:**
 - Gently place a piece of wood into the water basin.
 - **Observation:** Check whether the wood floats or sinks. Note whether it floats well or partially.
3. **Testing Foam:**
 - Add a piece of foam to the water basin next.
 - **Observation:** Observe how the foam behaves in the water. Determine if it floats easily and whether it stays afloat or sinks.
4. **Testing Plastic Bottle:**
 - Insert an empty plastic bottle into the water basin.
 - **Observation:** Monitor the behavior of the plastic bottle. Note if it floats and if it remains upright or moves around.
5. **Testing PVC Material:**
 - Finally, add the PVC material to the water basin.
 - **Observation:** Observe how the PVC interacts with the water. Check if it floats or sinks and whether it remains afloat easily.

What to Observe:

- **Floating Ability:** Does the material float or sink?
- **Stability:** Does the floating material remain stable and upright, or does it tip over?
- **Buoyancy:** How effectively does each material support its own weight in the water?

Results and Analysis:

After testing each material, compare your observations:

1. **Wood:**
 - **Floating Ability:** Typically floats because it is less dense than water.
 - **Stability:** Can be stable, but may tip over if not balanced.
 - **Buoyancy:** Good buoyancy; effectively supports its weight.
2. **Foam:**
 - **Floating Ability:** Floats very well due to its low density.
 - **Stability:** Stays upright and stable, making it ideal for flotation.
 - **Buoyancy:** Excellent buoyancy; supports significant weight without sinking.
3. **Plastic Bottle:**
 - **Floating Ability:** Floats easily when empty.
 - **Stability:** Can remain upright but may tip over depending on water movement.
 - **Buoyancy:** Good buoyancy; effectively supports its weight.
4. **PVC:**
 - **Floating Ability:** Generally floats, but buoyancy can vary based on density.
 - **Stability:** Often stable and upright if designed correctly, but shape can affect this.
 - **Buoyancy:** Generally good buoyancy; supports weight but may not be as effective as foam or wood.

Conclusion:

- **Best Floating Material:** Foam floats the best due to its low density and excellent buoyancy.
- **Most Stable Material:** Foam is typically the most stable and remains upright easily.
- **Best Buoyancy:** Foam also demonstrates the best buoyancy, efficiently supporting weight without sinking.

Now it's your turn to decide which material are you going to use?

Let the students answer this question by themselves according to the results of the experiment.

EDP

Return back to the EDP paper and add the materials they are going to use to the plan part.



UNIT 01 [Beginner]

Chapter 01: Mechanics

LESSON 05: Best Design for Your Floater.

Time frame: 90 mins

Story

Apply the same strategies used in Part 1 of the story.

Think

1. What Will Make the Floater Float Well?

Possible Solutions:

- **Use Light Materials:** Choose materials that are light and can float, like foam or empty plastic bottles.
- **Good Shape:** Make the floater wide and flat at the bottom to help it stay balanced and not tip over.
- **Add Buoyancy:** Include items that can hold air, like sealed containers, to help it float better.
- **Even Weight:** Make sure the weight on the floater is spread out evenly so it doesn't sink on one side.

2. How Much Weight Can It Hold?

Possible Solutions:

- **Weight Testing:** Students can add small weights (like coins or washers) to the floater one by one and see how much it can hold before it sinks.
- **Understanding Buoyancy:** Explain that when the floater displaces water, it can hold a certain weight without sinking, and this is called buoyancy.
- **Material Strength:** Talk about how strong the materials are and how that affects how much weight the floater can carry.

Explore

Do it

Materials Needed:

- **Modeling Materials:** Use materials like cardboard, foam, or plastic sheets for building prototypes. For a more hands-on approach, you might use blocks of wood or hollow logs if you have access.
- **Weights:** Small weights or objects that can be added to test buoyancy.
- **Bathtub or Large Water Container:** To test the floaters.
- **Measuring Tools:** Ruler or measuring tape for accurate sizing.
- **Design Tools:** Pencils, markers, or software for sketching designs (optional).

Steps to Follow:

1. **Create Models of Different Sizes:**
 - **Design Prototypes:** Using your chosen materials (cardboard, foam, plastic), create several models of floaters with the same shape but different sizes. For example, if you're designing a rectangular floater, make a few with varying lengths, widths, and heights.
 - **Mirror Images:** Ensure that each model is symmetrical or has mirror image sides. This helps in assessing how balanced the floater is.
2. **Design Considerations:**
 - **Solid vs. Hollow:** Consider the difference between a solid block (like a thick piece of wood) and a hollow structure (like a hollow log or a boat hull). A hollow design often provides better buoyancy.
 - **Adjust Designs:** Experiment with different designs. For example, you might compare a solid block with a hollowed-out version of the same size to see how the shape affects floating.
3. **Test for Balance and Buoyancy:**
 - **Add Weights:** Gradually add different weights to your floaters to test how they handle extra load. This will help determine the best balance between floating high and staying steady.
 - **Observe Performance:** Note how each design performs with added weights. Does it float high or does it sink slightly? Is it stable or does it tip over easily?
4. **Test in a Bathtub:**
 - **Place Floaters in Water:** Fill a bathtub or large container with water and place each floater model in it.

- **Observation:** Watch how each model behaves in the water. Check for stability, how much of the floater is above the water, and how it handles additional weights.

Write Your Observations:

- **Floating Height:** Record how high or low each model floats. Does a larger model float higher than a smaller one?
- **Stability:** Note if the floater remains steady or tilts. How does it react to different weights?
- **Design Effectiveness:** Compare the solid versus hollow designs. Which provides better buoyancy and stability?

Write Your Conclusion:

- **Best Design:** Based on your observations, determine which design is the best. Was a certain size or shape more effective? Did the hollow or solid model perform better?
- **Balance and Buoyancy:** Summarize the balance between floating high and staying steady. Which model achieved the best balance?
- **Recommendations:** Suggest any improvements or modifications that could enhance the floater's performance. For example, would adjusting the size or shape improve stability?

This process helps you understand how different designs and materials affect buoyancy and stability, providing a practical approach to creating effective floaters.

Watch it

<https://www.youtube.com/watch>

1. Previewing the Video

- **Activity:** Before starting the video, discuss what buoyancy means and ask students what they already know about floating and sinking. Introduce any key vocabulary they will encounter in the video.
- **Purpose:** This sets the context and prepares students to look for specific information while watching.

2. Guided Viewing

- **Activity:** Play the video in short segments, pausing at key moments to ask questions and facilitate discussions. For example, pause after demonstrations of buoyancy and ask:
 - What did you see happening?
 - Why do you think that object floated or sank?
- **Purpose:** This encourages students to think critically about what they are watching and helps them process the information.

3. Interactive Note-Taking

- **Activity:** Provide students with a graphic organizer or note-taking guide where they can jot down important points, examples of buoyant and non-buoyant objects, and any questions they have.
- **Purpose:** This keeps students engaged and helps them organize their thoughts for later discussion.

4. Real-World Connections

- **Activity:** After watching, ask students to think about real-world examples of buoyancy they've seen, such as boats, life jackets, or floating toys. Discuss how understanding buoyancy is important in these contexts.
- **Purpose:** Connecting the video content to their everyday experiences enhances understanding and relevance.

5. Discussion Questions

- **Activity:** After viewing, lead a discussion using open-ended questions. For example:
 - What surprised you in the video?
 - Can you think of other examples of buoyancy in nature?
 - How does buoyancy affect our daily lives?
- **Purpose:** This encourages deeper thinking and allows students to express their ideas and insights.

6. Post-Viewing Reflection

- **Activity:** Have students write a short reflection or summary of what they learned from the video. They can include a drawing of an object they think would float or sink and explain why.
- **Purpose:** Writing helps reinforce learning and allows students to articulate their understanding.

EDP

Return back to the EDP paper and add the design you're going to make in create part.



Show Case

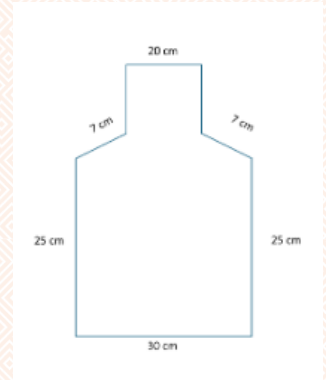
<https://youtu.be/kqCRbDLWMh0?si=Xv7CvKnyDQ7rk0W5>

Step 1: The boat body

Sunboard Instructions:

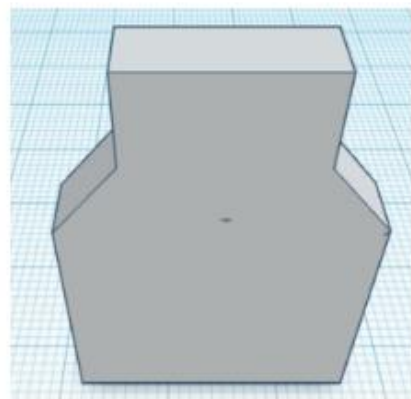
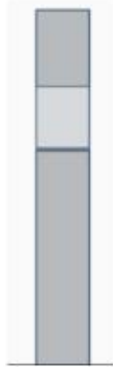
1. Cut the Foam Shapes

- Cut a rectangle of foam measuring 40cm in width and 30cm in length.
- Then cut the top part of it as shown in a picture.
- Cut out another identical shape.
- Prepare the following rectangles:
 - Two rectangles measuring 7cm in width and 3cm in height.
 - Two rectangles measuring 25cm in width and 3cm in height.
 - One rectangle measuring 30cm in width and 3cm in height.
 - One rectangle measuring 20cm in width and 3cm in height.



2. Assemble the Sunboard

- Use a glue gun or other suitable adhesive to attach the shapes together according to the provided image.



UNIT 01 [Beginner]

Chapter 02: Electricity

LESSON 01: Navigate Your Floater.

Time frame: 90 mins

Stepping Stone.

You need to know ..	You will be able to..
How electricity works.	Identify voltage and current how to measure them.
Basic circuit components (LED- Resistor- Battery- Switch).	Create more complicated circuit on a breadboard.
Connecting circuits on smart pad, and breadboard.	Control the speed of a motor.
Understanding of series and parallel circuits.	

Story

Apply the same strategies used in Part 1 of the story.

Think

Think of how series and parallel connections efficient light bulb brightness. This knowledge will help design a safe, efficient classroom lighting system using LEDs, a smart pad, and resistors.

Procedures:

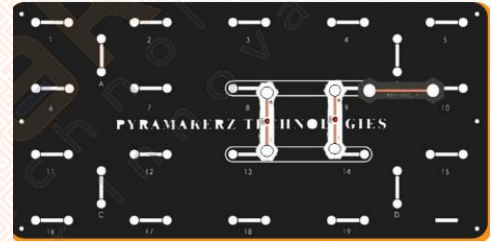
- 1- Divide students into groups of 3 to 5 students.
- 2- Distribute the smartpads.



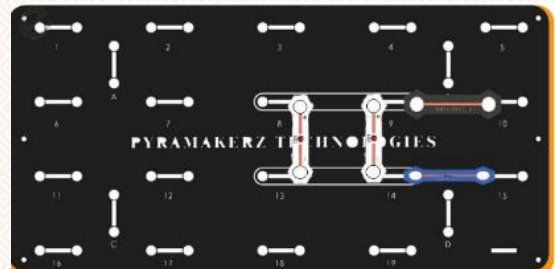
- 3- Give the needed electric components (LEDs- Wires- Resistors)
- 4- Set a specific time for them to apply the circuit.

Key Guide for the teachers..

- 1- Connect a wire between the battery's positive end and node 9.
- 2- Connect the first and second LED parallel.



- 3- Connect the two LED's negative end (The shorter end) to resistor's terminal in node 14 and connect the other resistor's to battery's negative end.
- 4- Connect smart pad with power source and see what happens.

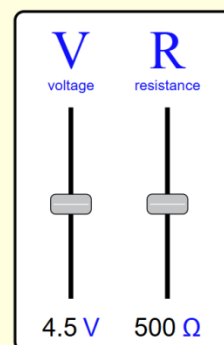
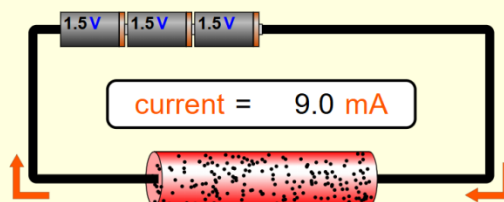


Explore

Do It

https://phet.colorado.edu/sims/html/ohms-law/latest/ohms-law_all.html

$$V = IR$$



Attribute Windows

Title: Exploring Voltage and Current

Objective: Students will learn the difference between voltage and current, how they work together, and how they power various devices.

Materials Needed: Open the simulation.

- **Instructions:**

- Divide students into small groups.
 - Ask each group to set up circuits using batteries of different voltages and observe the changes.
 - If using a multimeter, demonstrate how to measure voltage and current, then have students record their readings.
 - Lead a discussion about how voltage (battery strength) affects the brightness and current (the flow of electricity). Emphasize that higher voltage gives a stronger "push," which increases current and can make the bulb brighter.
- **Strategy: Inquiry-Based Learning** is effective here. Pose guiding questions: *What do you notice about the reading of the current when you use different batteries?* This encourages students to make observations and form conclusions.
 - **Strategy:** Use a **Conceptual Discussion** strategy here. Encourage students to share their group's observations and discuss why they think different batteries had different effects.

Watch it..

You may apply the same strategies previously used in this section.

Read it..

You may apply the same strategies previously used in this section.

Assessment

Focus

Activity: Building a Simple Circuit with a Motor and Switch

Objective: Students will learn how to create a basic circuit, understand motor polarity, and observe the effect of reversing terminals.

Materials Needed:

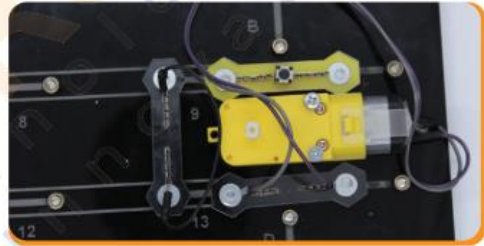
Smart pad – Motor- Switch.

STEPS

1 Connect the motor.



2 Connect the motor with a push button and a wire to close the circuit.



NOTE

Try to reverse the terminals of the motor and see what will happen!!

Students should observe that the shaft of the motor will rotate in the opposite direction.

Practice

- Building a Dual Motor Circuit on a Breadboard

Materials Needed:

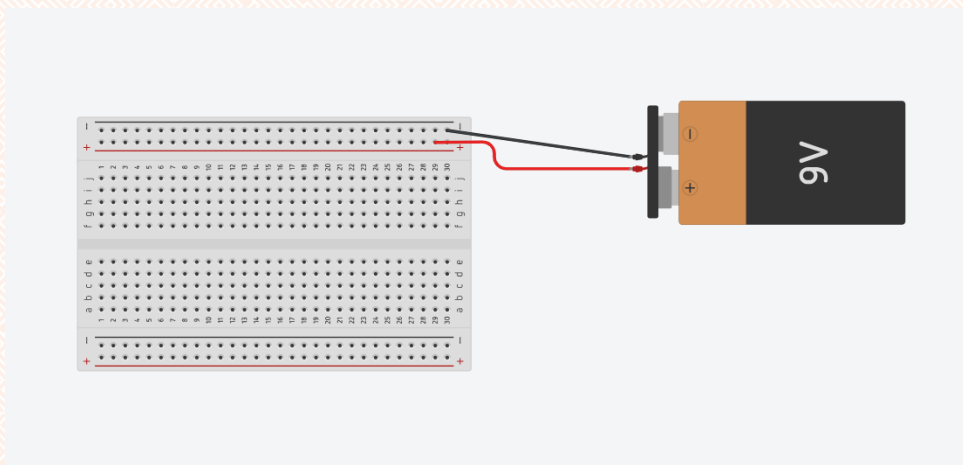
- Breadboard
- 2 small DC motors
- Wires (with stripped ends or jumper wires)
- Battery holder with batteries (matching the motor's voltage)

Step 1: Setting Up the Breadboard and Components

1. **Place the Breadboard and Components:** Set the breadboard on a flat surface. Place the battery holder and motors nearby.
2. **Identify Breadboard Sections:** Explain to students how the power rails (horizontal rows) and terminal strips (vertical columns) work on the breadboard.

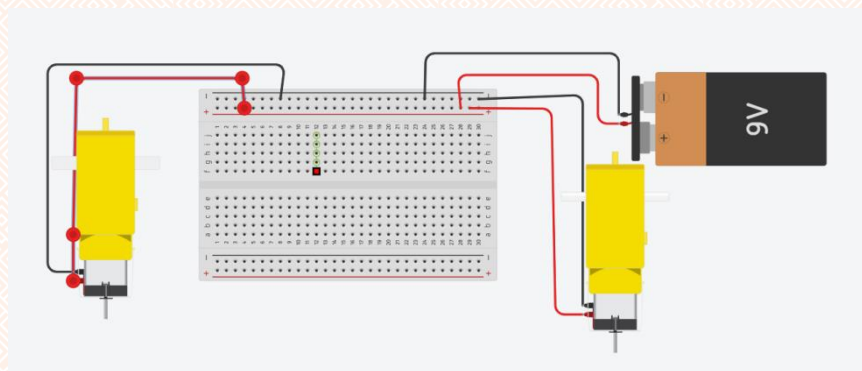
Step 2: Connecting the Power Source to the Breadboard

1. **Connect the Battery Holder to the Breadboard:** Insert the positive (red) and negative (black) wires from the battery holder into the power rails on the breadboard. Connect the red wire to the positive rail and the black wire to the negative rail.



Step 3: Adding the Motors

1. **Position the Motors on the Breadboard:** Place each motor so that its terminals are in different rows within the same section of the breadboard.
2. **Connect the First Motor in Parallel:**
 - Use a jumper wire to connect one terminal of the first motor to the positive power rail.
 - Connect the other terminal of the motor to the negative power rail.
3. **Connect the Second Motor in Parallel:**
 - Insert a jumper wire to connect the positive terminal of the second motor to the same row connected to the positive rail.
 - Connect the negative terminal of the second motor to the negative power rail.



Step 4: Testing the Circuit

Step 5: Reversing the Terminals of One Motor

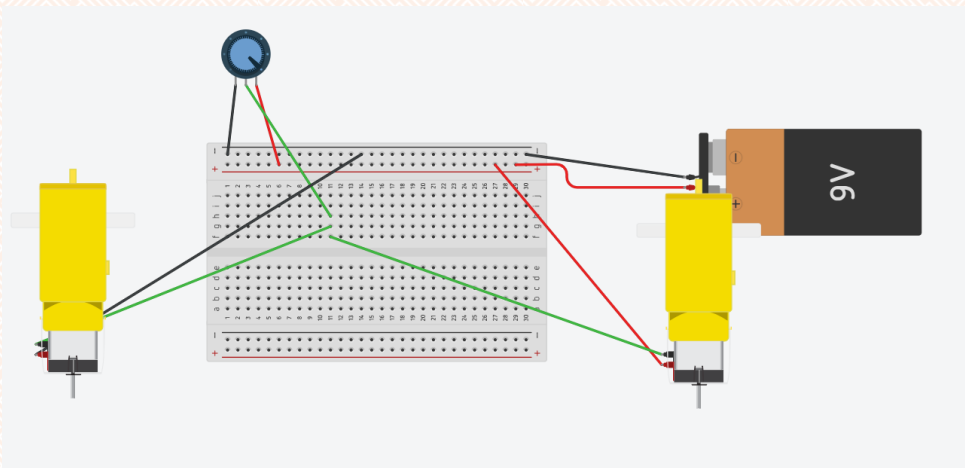
1. **Disconnect the Power:** Remove one of the battery holder connections from the power rail to cut off the power.
2. **Reverse One Motor's Connections:** Swap the wires connected to one motor, switching its positive and negative connections on the breadboard.
3. **Reconnect the Power:** Insert the battery connection back to the power rail and observe the motors. One motor should now spin in the opposite direction.

Challenge

This activity is an optional extension, intended to be introduced to students as a bonus task they can work on independently if they wish. It is not required to be completed during the session.

Answer:

- 1- Increase the number of batteries.
- 2- Add potentiometer.



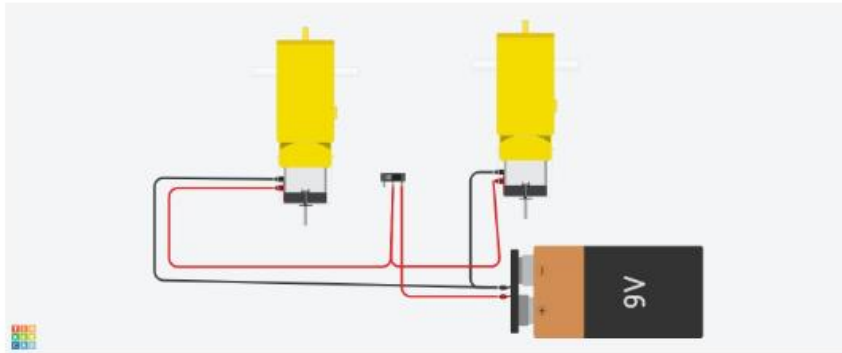
EDP

Return back to the EDP paper and add the materials they are going to use to the create part.



Showcase

Adding motors and wheels to the floater.



Materials used:

- 2 dc motor yellow
- 9v battery
- Switch

UNIT 01 [Beginner]

Chapter 03: Design Domain.

LESSON 01: Let's Dive in.

Time frame: 90 mins

Stepping Stone

You should know ...	You should be able to...
Tinker cad workspace including the design grid, toolbars, and navigation controls.	Understand what a propeller looks like and how it functions.
The primary functions such as move, scale, rotate, and duplicate and access and use basic shapes.	Create and shape the individual blades of the propeller using basic shapes like cylinders or elongated cones.
How to combine multiple shapes into one object and how to separate them.	Connect propeller to motor

Story

Apply the same strategies used in Part 1 of the story.

Think



What basic shapes we might use for the blades of our propellers?

Strategy for Teachers: Guided Inquiry and Exploration

1. **Start with Open-Ended Questioning:** Begin by asking, *"What basic shapes do you think we might use for the blades of our propellers?"* Allow students to brainstorm and discuss possible shapes without focusing on right or wrong answers. Encourage them to think about basic geometric shapes (rectangles, triangles, ellipses, etc.).
2. **Decompose the Problem:** Guide students to decompose the design of a propeller blade by identifying what features they think are essential. Ask follow-up questions, such as:
 - *"What do you think the blade needs to do when it spins?"*
 - *"How do you think shape affects the way air moves around the blade?"*

This helps students focus on the purpose of each part and how it contributes to the function of the whole propeller.

3. **Abstract Key Elements:**

Once students have decomposed the design, prompt them to consider the most important features across different shapes. Ask questions like:

- *"What parts of the shape might be necessary to create lift or push air?"*
- *"Which shapes seem most effective for moving air?"*

Here, students start to generalize, or abstract, the core features of a blade shape that are necessary for effective air movement (e.g., broad surface area, slanted edges).

4. **Experiment with Shape Prototypes:** If possible, allow students to create simple propeller blade prototypes with different shapes. This can be done with paper, cardboard, or other materials. Encourage students to observe how the shape affects movement when spun manually or with a small fan.

Explore

Watch it...

You may apply the same strategies previously used in this section.

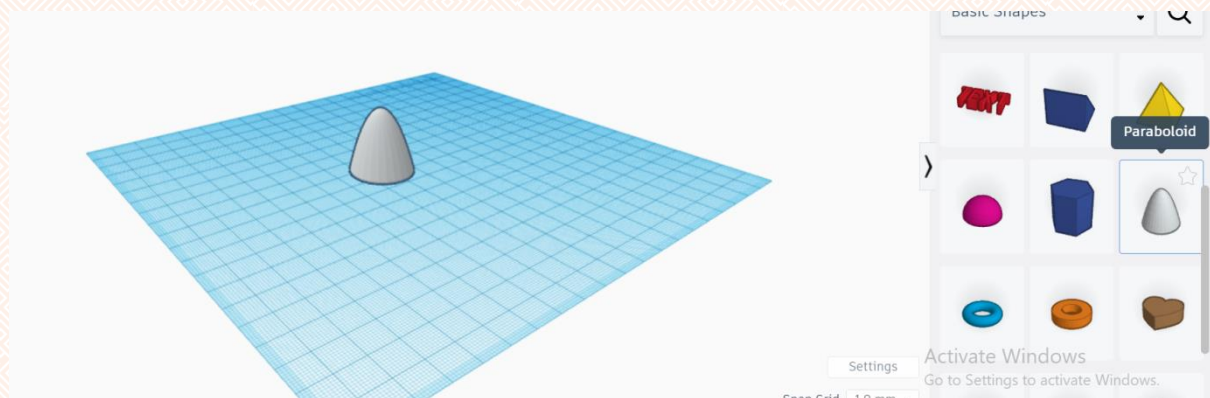
Do it

Designing the Propeller in Tinkercad:

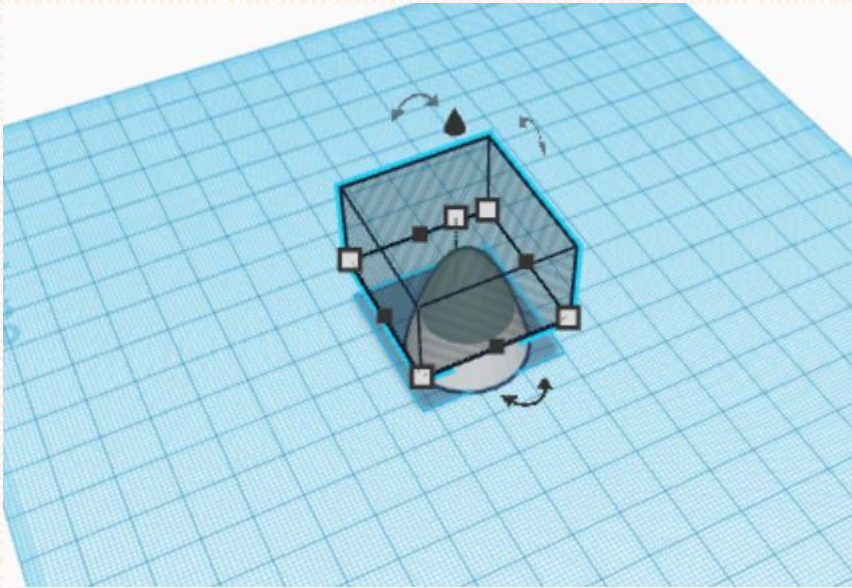
Using the abstraction and decomposition skills they developed in the thinking question, have students create their propeller design in Tinkercad.

Steps:

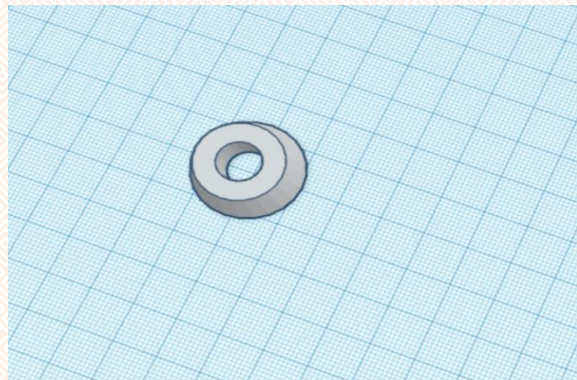
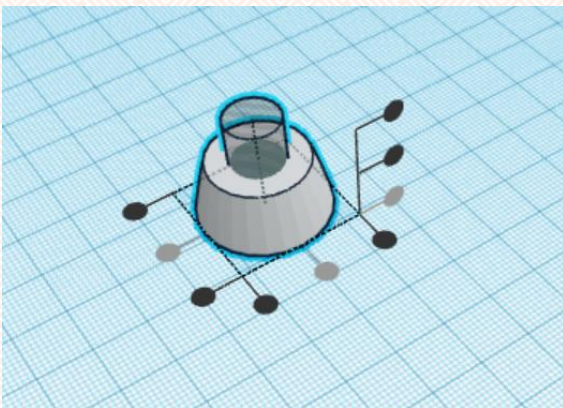
- 1- Choose the paraboloid shape and set its dimensions to $L=25$ $W=25$ $h=26$



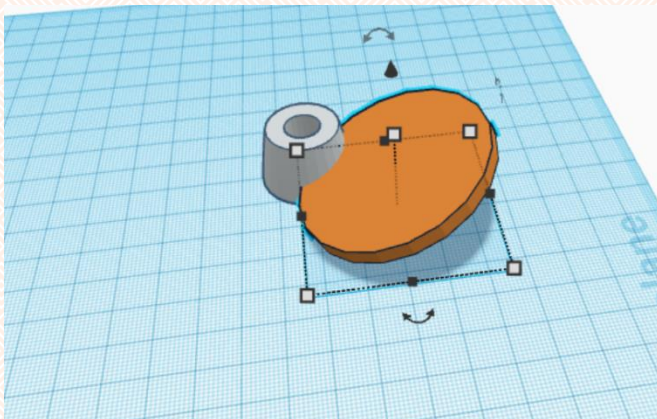
2- Add a hole box at the top and group them to cut the edge.



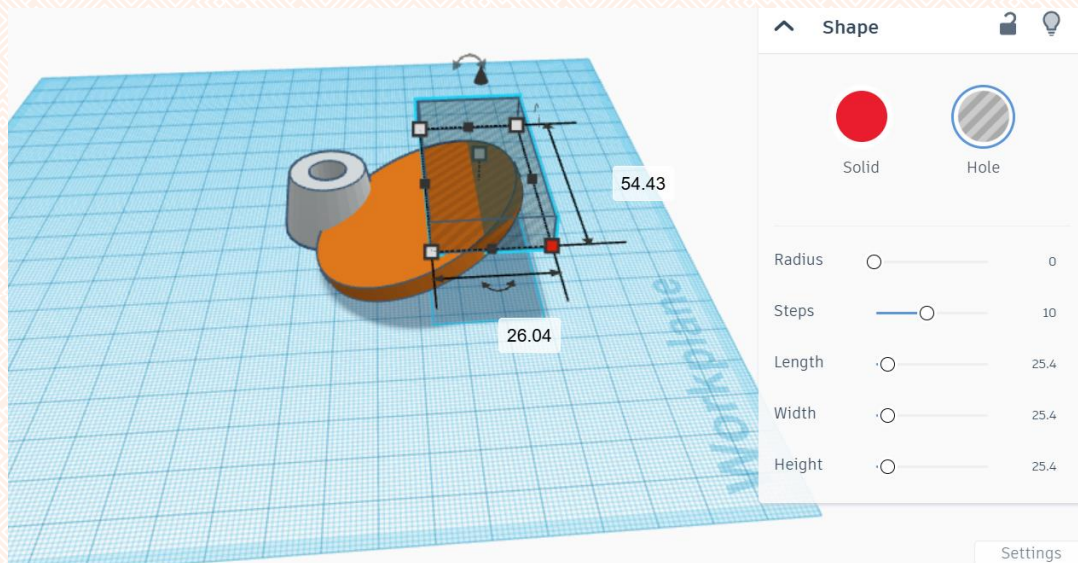
3- Add a hole cylinder inside it, make its dimensions L= 10 W= 10 H=20 then group it to make a hole in the propeller.



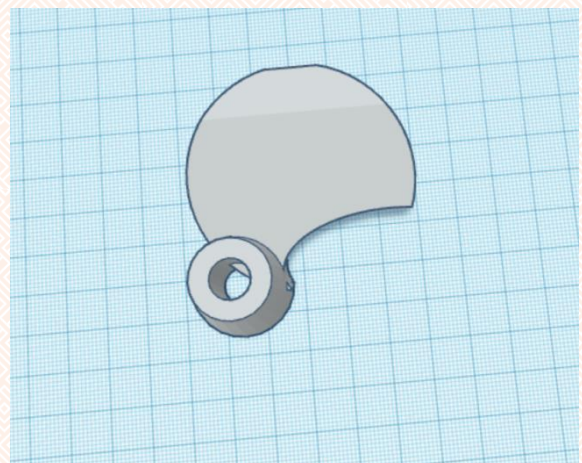
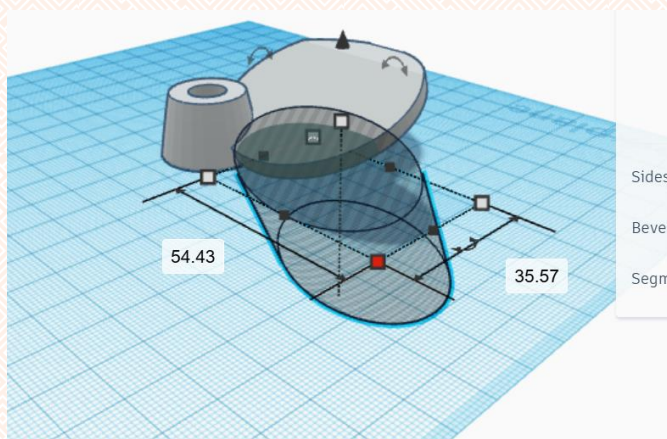
4- Now add a cylinder it's dimensions, L=53 , W= 53 and H= 6 and make it sylinted to the right.



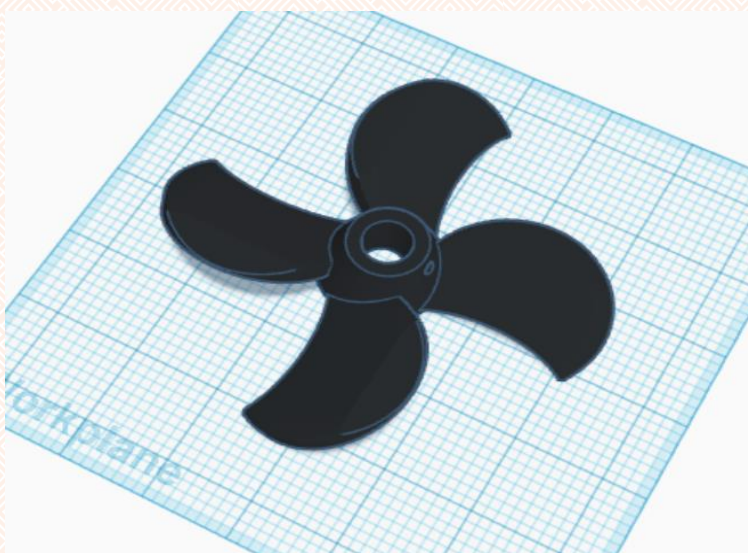
5- Now add a hole box and make its dimensions like the image and group it.



6- Now make a cylinder its dimensions as shown in the image and add it to make a hole by grouping them,.



7- Repeat the steps again to make all the shafts.



Assessment

Focus

What is a key factor in designing a propeller for the Flouter?

- **Answer:** b) Number of blades

How does the number of blades affect a propeller's performance?

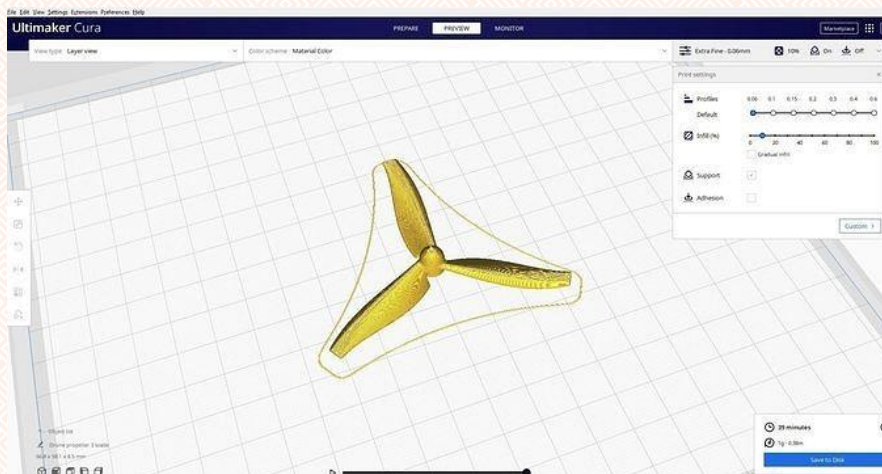
- **Answer:** c) The number of blades can affect stability and thrust

Which blade shape might reduce drag and increase efficiency?

- **Answer:** b) Curved

Practice

Make the students open cura, adjust the shape and print it on the 3D printer.



Challenge

This activity is an optional extension, intended to be introduced to students as a bonus task they can work on independently if they wish. It is not required to be completed during the session.

EDP

Return back to the EDP paper and add the materials they are going to use to the create part.



Show Case

Attach the propeller to your model.



UNIT 02 [Intermediate]

Chapter 01: Mechanics Domain.

LESSON 01: Let's move our trash up.

Time frame: 45 mins

Stepping Stone

You should know ...	You should be able to...
Basic knowledge of gears and how they interact.	Understanding of mechanism of working of conveyor belt.
Definition and some types of forces such as friction force.	
Identify pulley	

Story

Apply the same strategies used in Part 1 of the story.

Think

Apply the same strategies used before in this section.

1. What parts are needed to build a conveyor belt?

- **Solution:** A basic conveyor belt requires:
 - **Belt Material:** Often a flat or flexible material (e.g., rubber or canvas) that moves along the conveyor's frame.
 - **Rollers:** Two or more rollers support the belt and help move it along. The rollers are positioned at both ends of the conveyor.
 - **Motor:** Powers the belt, usually connected to one of the rollers to drive the movement.
 - **Frame/Structure:** A sturdy frame to hold the rollers and support the belt and objects.
 - **Power Source:** Often an electric power source connected to the motor to keep it running.

2. How do these parts work together to move objects?

- **Solution:**
 - **Motor and Roller Connection:** The motor powers one of the rollers, known as the drive roller, which turns and moves the belt. The other roller acts as an idler to keep the belt aligned.
 - **Belt and Roller Interaction:** As the motor-driven roller rotates, it pulls the belt along with it, creating a continuous loop.
 - **Object Transport:** Objects placed on the belt are carried forward due to the belt's movement, which is guided by the motor and rollers.

3. Does friction play a role in this mechanism?

- **Solution:**
 - **Between Belt and Rollers:** Yes, friction between the belt and rollers is essential, as it enables the belt to grip the rollers and move smoothly without slipping.
 - **Between Belt and Objects:** Friction between the belt and the objects on it is also important; it prevents objects from sliding off and ensures they move along with the belt.
 - **Control and Efficiency:** A balance of friction is crucial. Too little friction may cause slipping, while too much can lead to wear on the belt and inefficient operation.

Explore

Do it...

<https://www.youtube.com/watch?v=i9V3rPQ0ZW4>

Follow the steps in the video to make the activity with the students.

Observation for the Conveyor Belt Race Game Activity:

1. **Using a Rough Belt:**
 - **Observation:** Objects on the rough belt move more slowly, as there is increased friction between the belt and the objects. The rough surface provides more grip, making it harder for the objects to slide quickly, which slows down their movement.
2. **Using a Smooth Belt:**
 - **Observation:** Objects on the smooth belt move faster because there is less friction between the belt and the objects. The smoother surface allows the objects to slide more easily, leading to faster movement.

Watch it...

You may apply the same strategies previously used in this section.

EDP

Return back to the EDP paper and add the materials they are going to use to the plan part.

Plan

Assessment

Focus

- what **part of the belt system is like a smooth track?**
 - **Answer:** C. Belt Support
- which **part of the belt system can change the direction of the belt?**
 - **Answer:** C. Pulley
- what **is the main function of the Drive Unit?**
 - **Answer:** B. To power the belt

Practice

<https://www.youtube.com/watch?v=8YzTE8jBEU0&t=172s&pp=ygURY29udmV5b3lgYmVsdCBkaVk%3D>



Follow the video steps to make the conveyor belt.

Challeng

This activity is an optional extension, intended to be introduced to students as a bonus task they can work on independently if they wish. It is not required to be completed during the session.

EDP

Return back to the EDP paper and add the materials they are going to use to the create part.



Show Case

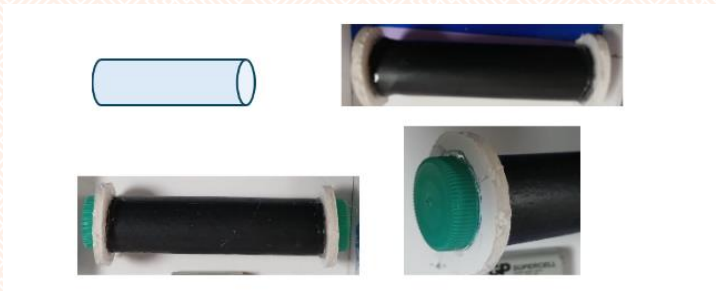
Step 3: Building the Conveyor belt:

1. Cut Foam Pieces:

- Cut four square pieces of foam, each measuring 4cm by 4cm.
- Cut two rectangular pieces of foam, each measuring 4.5cm in length and 30-35cm in width.
- Cut one rectangular piece of foam measuring 4.5cm in length and 20cm in width.
- Cut four circles out of foam, each with a diameter of 5cm.

2. Prepare PVC Pipes:

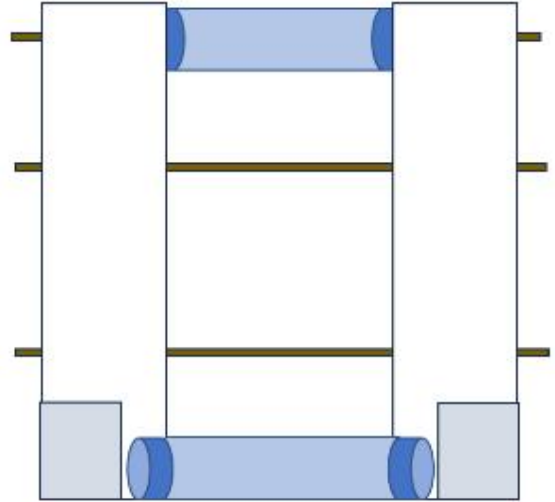
- Cut two pieces of PVC pipe, each with a length of 13cm.
- Attach a foam circle to each end of both PVC pipes.
- Secure a bottle cap on top of each circle.



3. Assemble the Conveyor Frame:

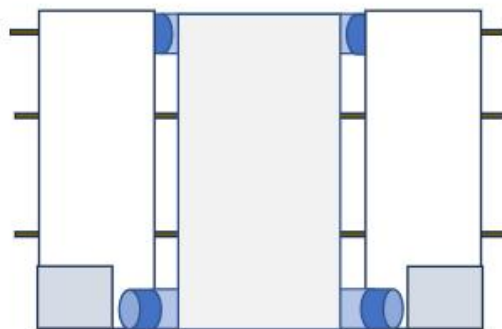
- Use glue to attach the four square foam pieces to the ends of the two long rectangular foam pieces, forming a rectangular frame.
- Position the two PVC pipes with attached circles and bottle caps between the two long foam pieces.

- Use three sticks to reinforce the structure: two sticks between the long foam pieces and one stick passing through the PVC pipes and connecting to the long foam



4. **Create the Conveyor Belt:**

- Wrap fabric around the two PVC pipes to create a continuous belt. Secure the fabric in place with glue.
- Attach a motor to each of the two square foam pieces at opposite ends of the conveyor belt.
- Create holes in the bottle caps to accommodate the motor shafts.



Note: Don't attach the motors in this domain

UNIT 02 [Intermediate]

Chapter 02: Electric Domain.

LESSON 01: Function Your Conveyor belt.

Time frame: 45 mins

Stepping Stone

You should know ...

Understanding of simple circuits and components (batteries , switches , motors)
experience with connecting components on a breadboard or using connectors

You should be able to...

Design and build separate circuits to control multiple motors
Manage power distribution using separate batteries for each motor ensuring stable and efficient power supply to each component

Story

Apply the same strategies used in Part 1 of the story.

Think

Apply the same strategies used before in this section.

Question 1: How can we connect the motor of the conveyor belt separately?

- **Possible Answers:**

- "We could use a separate power source or battery just for that motor, so it runs independently."
- "We could connect the motor to its own switch, allowing us to turn it on or off without affecting other parts."
- "If we have a controller or a circuit board, we could give each motor its own connection point so they work separately."

Question 2: What challenges might we face in controlling multiple motors independently?

- **Possible Answers:**

- "It might be hard to keep the motors synchronized because they could run at different speeds if they're powered separately."
- "If we have too many switches or controllers, it could get confusing to control each motor without accidentally turning off the wrong one."
- "The power supply might get overloaded if we try to run too many motors at the same time, especially if each has its own connection."

Explore

Do it...

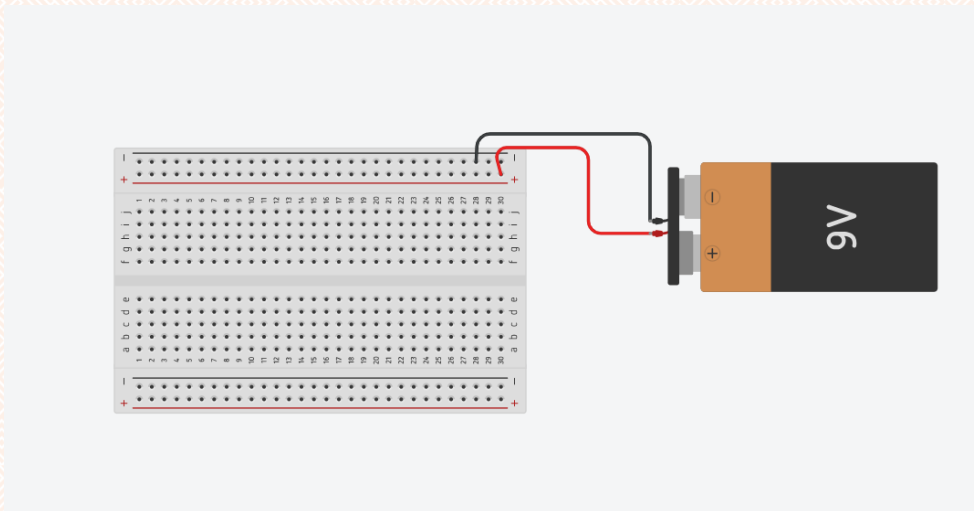
Connect a motor and a switch on a breadboard to control its operation.

Materials Needed:

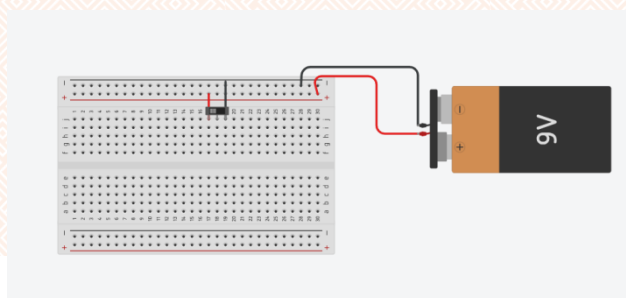
- Breadboard
- DC motor
- Switch
- Jumper wires
- Power source (e.g., a 9V battery with a battery clip)

Instructions:

- 1. Place the Motor on the Breadboard:**
 - Position the motor so its terminals are accessible on the breadboard, making sure it is secure.
- 2. Connect the Power Source:**
 - Attach the positive wire from the battery to the positive rail on the breadboard.
 - Connect the negative wire from the battery to the negative rail on the breadboard.

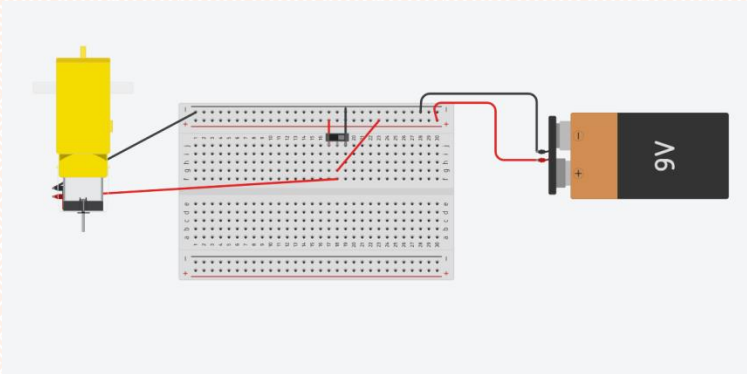


- 3. Insert the Switch:**
 - Place the switch onto the breadboard, ideally in a way that each pin is in its own separate row for easy connections.



4. Connect the Motor to the Switch:

- Use a jumper wire to connect one terminal of the motor to one pin on the switch.



5. Test the Circuit:

- Turn on the switch to test if the motor starts running. If connected correctly, flipping the switch should control the motor's operation, turning it on or off.

Watch it...

You may apply the same strategies previously used in this section.

Assessment Exercise

Focus

Why do motors need a lot of power to work?

A. Because they have to move large, heavy objects

Practice

Allow students to choose the type of switch they want to use and explain their reasoning.

Show Case

Now add the motors from the last part to the conveyor belt.

UNIT 02 [Intermediate]

Chapter 03: Robotics Domain.

LESSON 01: Let's Control The Floater.

Time frame: 45 mins

Stepping Stone

You should know ...	You should be able to...
What an Arduino is and its basic components.	Writing and understanding more complex code to control motors.
How to upload code to an Arduino.	Understanding of motor drivers and how to use them.
Familiarity with the Arduino IDE and basic syntax.	
How to connect components on a breadboard.	

Story

Apply the same strategies used in Part 1 of the story.

Think

Why is it necessary to consider the voltage and current requirements of motors when integrating them with control circuitry like an Arduino?

- **Protecting the Arduino and Motor:**

- "If the motor needs more power than the Arduino can handle, it could break parts of the circuit. Making sure the motor and Arduino match in power helps protect them."

- **Giving the Motor Enough Power:**

- "Motors need a lot of power to run. If we don't give the motor the right amount of power, it might not work well or could get too hot. Checking the power needs makes sure the motor runs smoothly."

- **Keeping the Motor Running Steady:**

- "If the power isn't right, the motor might not run at a steady speed or could stop suddenly. The right power helps it run smoothly and predictably."

- **Making the Motor and Arduino Last Longer:**

- "Using the right amount of power helps the motor and Arduino last longer. Too much or too little power could wear them out faster."

- **Avoiding Electrical Problems:**

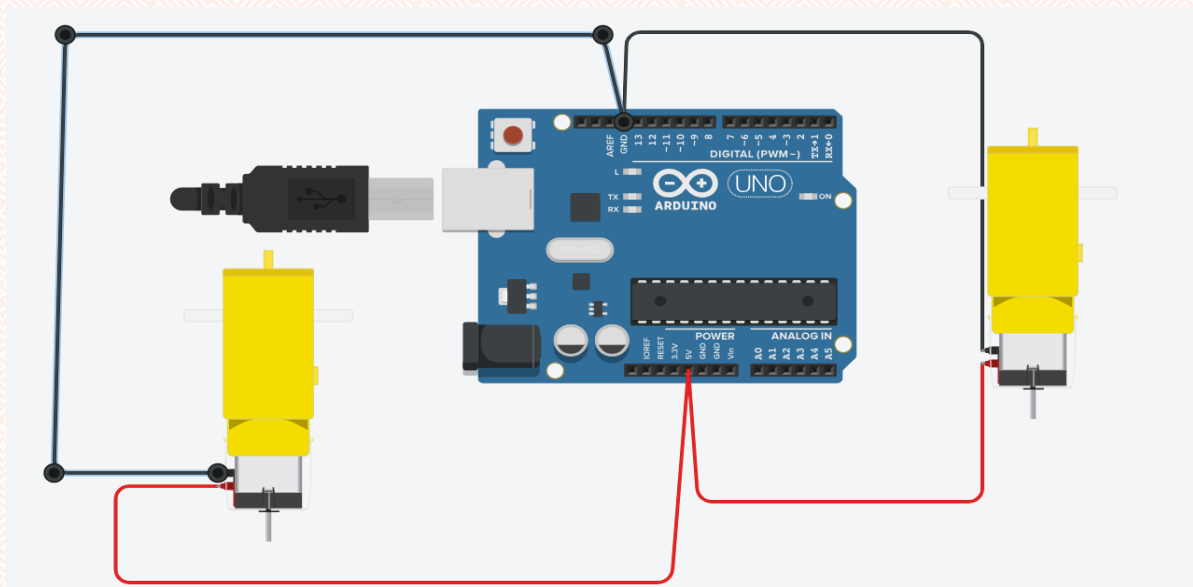
- "Motors can cause little electrical surges that can mess with the Arduino. Using the right power helps prevent these surges from causing problems."

- **Getting the Right Speed and Strength:**

- "If we match the power correctly, the motor will work at the right speed and strength. This is important if we want it to move things in a certain way."

Explore

Do it...



Materials Needed:

1. **Arduino Board:** Any Arduino board (e.g., Arduino Uno).
2. **Motors:** Two DC motors (e.g., 6V or 12V motors).
3. **Motor Driver Module:** Such as the L298N or L293D motor driver.
4. **Breadboard:** For connecting components.
5. **Jumper Wires:** To connect the Arduino to the motors and motor driver.
6. **Power Source:** Battery or external power supply (appropriate for your motors).
7. **Resistors:** If needed for current-limiting (depending on the motor driver).

8. **Arduino IDE:** Software installed on your computer to write and upload code to the Arduino.

Steps:

1. Connecting Motors Directly to the Arduino:

1. Set Up the Arduino and Motors:

- **Connect the Motors:**
 - Connect the positive terminal of the first motor to one of the Arduino's digital output pins (e.g., Pin 9).
 - Connect the negative terminal of the first motor to the ground (GND) on the Arduino.
 - Repeat for the second motor, connecting it to a different digital output pin (e.g., Pin 10).

2. Write and Upload Code to Arduino:

- Open the Arduino IDE on your computer.
- Write a simple code to turn the motors on and off. Here's an example sketch:

cpp

Copy code

```
// Define motor pins
const int motor1Pin = 9;
const int motor2Pin = 10;

void setup() {
  // Set motor pins as outputs
  pinMode(motor1Pin, OUTPUT);
  pinMode(motor2Pin, OUTPUT);
}

void loop() {
  // Turn motors on
  digitalWrite(motor1Pin, HIGH);
  digitalWrite(motor2Pin, HIGH);
  delay(1000); // Wait for 1 second

  // Turn motors off
  digitalWrite(motor1Pin, LOW);
  digitalWrite(motor2Pin, LOW);
  delay(1000); // Wait for 1 second
}
```

3.
 - Upload the code to the Arduino board.
4. **Test the Circuit:**
 - Observe if the motors turn on and off according to the code. Note if they behave as expected.
5. **Write Your Observations:**
 - **Motor Operation:** Did the motors turn on and off correctly?
 - **Motor Performance:** Were the motors running smoothly or making any unusual sounds?

2. Connecting Motors Through a Motor Driver:

1. Set Up the Motor Driver:

- **Connect the Motor Driver to the Arduino:**
 - Connect the motor driver's input pins (IN1 and IN2 for motor 1; IN3 and IN4 for motor 2) to digital output pins on the Arduino (e.g., Pin 9, 10 for motor 1 and Pin 11, 12 for motor 2).
 - Connect the output pins of the motor driver to the terminals of the motors.
 - Connect the motor driver's power input (VCC and GND) to the external power source suitable for the motors. Connect the GND of the external power source to the GND of the Arduino to ensure a common ground.

2. Write and Upload Code to Arduino:

- Modify the previous code to control the motor driver instead of directly controlling the motors:

cpp

Copy code

```
// Define motor driver pins
const int motor1Pin1 = 9;
const int motor1Pin2 = 10;
const int motor2Pin1 = 11;
const int motor2Pin2 = 12;

void setup() {
  // Set motor driver pins as outputs
  pinMode(motor1Pin1, OUTPUT);
  pinMode(motor1Pin2, OUTPUT);
  pinMode(motor2Pin1, OUTPUT);
  pinMode(motor2Pin2, OUTPUT);
}

void loop() {
```



```

// Turn motors on
digitalWrite(motor1Pin1, HIGH);
digitalWrite(motor1Pin2, LOW);
digitalWrite(motor2Pin1, HIGH);
digitalWrite(motor2Pin2, LOW);
delay(1000); // Wait for 1 second

// Turn motors off
digitalWrite(motor1Pin1, LOW);
digitalWrite(motor1Pin2, LOW);
digitalWrite(motor2Pin1, LOW);
digitalWrite(motor2Pin2, LOW);
delay(1000); // Wait for 1 second
}

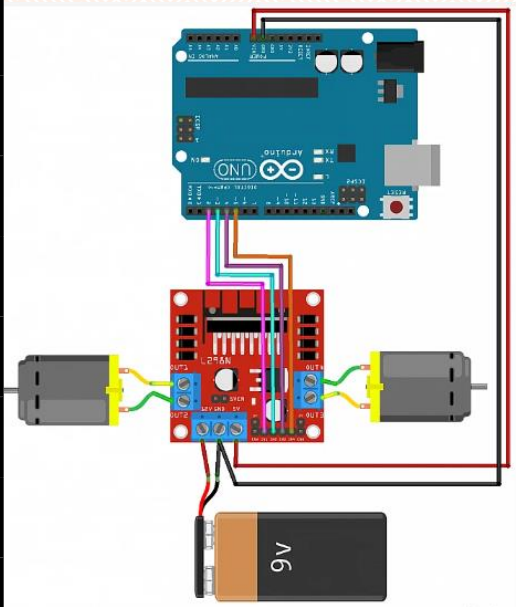
```

3.
 - Upload the modified code to the Arduino board.
4. **Test the Circuit:**
 - Observe how the motors behave when controlled through the motor driver.
5. **Write Your Observations:**
 - **Motor Operation:** Did the motors operate as expected when controlled by the motor driver?
 - **Performance Comparison:** How did the performance compare to when the motors were connected directly to the Arduino? Was there any difference in how smoothly the motors ran or how effectively they responded to the commands?

Conclusion:

- **Direct Connection:** When motors are connected directly to the Arduino, they are controlled directly by the microcontroller's output pins, which might not handle high currents well, potentially leading to limited performance or overheating issues.
- **Motor Driver:** Using a motor driver allows the Arduino to control higher currents needed for the motors and provides better performance and control. The motors should run more smoothly and be better protected from potential damage.

This experiment demonstrates how using a motor driver can improve the control and performance of motors in a circuit.



Watch it...

You may apply the same strategies previously used in this section.

Assessment Exercise

Focus

Question 1: What is the primary function of a motor driver in an Arduino project?

- **Answer: B) To control the speed and direction of motors**

Question 2: Which motor driver is commonly used with Arduino for controlling DC motors?

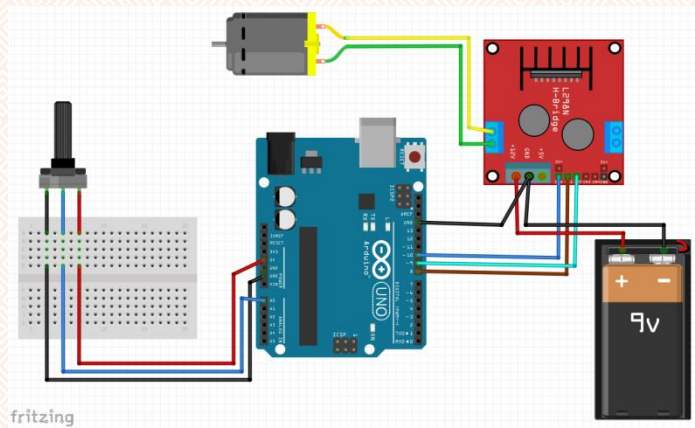
- **Answer: A) L298N**

Question 3: What type of motors can the L298N motor driver control?

- **Answer: C) DC motors and stepper motors**

Practice

Add the potentiometer to the circuit.



```
// Define pins
int potPin = A0;           // Potentiometer connected to analog pin A0
int motor1Pin1 = 9;        // Motor 1 IN1 connected to digital pin 9
int motor1Pin2 = 8;        // Motor 1 IN2 connected to digital pin 8
int motor2Pin1 = 7;        // Motor 2 IN3 connected to digital pin 7
int motor2Pin2 = 6;        // Motor 2 IN4 connected to digital pin 6
int enablePin1 = 10;       // Motor 1 enable pin (ENA) connected to pin 10
int enablePin2 = 5;        // Motor 2 enable pin (ENB) connected to pin 5
```

```
void setup() {
    // Set motor control pins as output
    pinMode(motor1Pin1, OUTPUT);
    pinMode(motor1Pin2, OUTPUT);
    pinMode(enablePin1, OUTPUT);

    pinMode(motor2Pin1, OUTPUT);
    pinMode(motor2Pin2, OUTPUT);
    pinMode(enablePin2, OUTPUT);

    // Start both motors in a specific direction
    digitalWrite(motor1Pin1, HIGH); // Motor 1 direction
    digitalWrite(motor1Pin2, LOW);

    digitalWrite(motor2Pin1, HIGH); // Motor 2 direction
    digitalWrite(motor2Pin2, LOW);
}
```

```

void loop() {
  // Read the potentiometer value (0 to 1023)
  int potValue = analogRead(potPin);

  // Map the potentiometer value to motor speed (0 to 255)
  int motorSpeed = map(potValue, 0, 1023, 0, 255);

  // Set the speed for both motors
  analogWrite(enablePin1, motorSpeed); // Motor 1 speed
  analogWrite(enablePin2, motorSpeed); // Motor 2 speed

  // Small delay for smoother operation
  delay(10);
}

```

Explanation of the Code

1. **Potentiometer Input:** The potentiometer is connected to analog pin A0, allowing the code to read the value and use it to control both motors' speed.
2. **Motor Control:** Each motor has two control pins to set its direction (`motor1Pin1`, `motor1Pin2` for Motor 1; `motor2Pin1`, `motor2Pin2` for Motor 2). Both motors are set to rotate in one direction for simplicity.
3. **Mapping:** The potentiometer's value (0–1023) is mapped to a speed range (0–255) using the `map()` function, allowing for smooth speed control based on the potentiometer's rotation.
4. **PWM Speed Control:** The `enablePin1` and `enablePin2` control the motor speeds using PWM (Pulse Width Modulation) to adjust the speed of each motor based on the potentiometer's position.

Here are some engaging strategies and ideas to make coding this Arduino project more fun and manageable for grade 5 students:

1. Break It Down into Small Steps

- **Step-by-Step Coding:** Instead of writing the entire code at once, break it down into small parts. For example, first, focus on just reading the potentiometer value. Once that's working, add the part that controls one motor, and finally, expand it to control both motors.
- **Explain Each Step:** After writing each small section of code, pause to discuss what it does. Ask the students questions like, "What do you think this line of code will do?" This approach helps them understand each part before moving on to the next.

2. Hands-On Coding Challenges

- **Coding Stations:** Set up different stations for each step (e.g., setting up the motor driver, reading the potentiometer, controlling the motor speed). This way, students can move from one station to the next, completing small sections of the code at each one.
- **Collaborative Coding:** Have students work in pairs or small groups. One student can write a few lines, and the other can check it. They switch roles after each section. Working together can make it feel less intimidating and more fun.

3. Gamify the Coding Process

- **Code and Test Race:** Turn coding and testing into a game by setting a small, manageable goal. For instance, challenge students to get the potentiometer reading printed on the screen first. Whoever finishes successfully gets a small reward or recognition.
- **Error-Spotting Game:** If something goes wrong with the code, let students play a “detective game” to find and fix the error. This approach makes debugging feel like a fun puzzle instead of a frustrating task.

4. Relate Coding to Real-World Examples

- **Everyday Connections:** Relate each section of code to real-world scenarios, such as how the potentiometer is like a volume knob or how PWM is like dimming a light. This helps students see why they're doing each step and makes the abstract concepts feel more concrete.
- **Show Real-World Motor Use:** Briefly explain that motors like these are in devices they use daily (fans, cars, robots) and that learning to code motors gives them power to control similar machines.

EDP

Return back to the EDP paper and add the materials they are going to use to the improve part.



Show Case

```
// Define motor pins

const int motor1Pin1 = 5; // IN1 on the L298N for Motor 1
const int motor1Pin2 = 6; // IN2 on the L298N for Motor 1
const int motor2Pin1 = 9; // IN3 on the L298N for Motor 2
const int motor2Pin2 = 10; // IN4 on the L298N for Motor 2


void setup() {

    // Set motor pins as outputs
    pinMode(motor1Pin1, OUTPUT);
    pinMode(motor1Pin2, OUTPUT);
    pinMode(motor2Pin1, OUTPUT);
    pinMode(motor2Pin2, OUTPUT);
}


void loop() {

    // Move forward for 30 seconds (30,000 ms)
    moveForward();
    delay(30000); // 30 seconds

    // Turn right for 30 seconds
    turnRight();
    delay(30000); // 30 seconds

    // Move backward for 30 seconds
    moveBackward();
    delay(30000); // 30 seconds
```



```
// Turn left for 30 seconds  
turnLeft();  
delay(30000); // 30 seconds  
}  
  
// Functions to control the directions of the motors  
void moveForward() {  
    digitalWrite(motor1Pin1, HIGH);  
    digitalWrite(motor1Pin2, LOW);  
    digitalWrite(motor2Pin1, HIGH);  
    digitalWrite(motor2Pin2, LOW);  
}  
  
void moveBackward() {  
    digitalWrite(motor1Pin1, LOW);  
    digitalWrite(motor1Pin2, HIGH);  
    digitalWrite(motor2Pin1, LOW);  
    digitalWrite(motor2Pin2, HIGH);  
}  
  
void turnRight() {  
    digitalWrite(motor1Pin1, HIGH);  
    digitalWrite(motor1Pin2, LOW);  
    digitalWrite(motor2Pin1, LOW);  
    digitalWrite(motor2Pin2, HIGH);  
}
```

```
void turnLeft() {  
  
    digitalWrite(motor1Pin1, LOW);  
  
    digitalWrite(motor1Pin2, HIGH);  
  
    digitalWrite(motor2Pin1, HIGH);  
  
    digitalWrite(motor2Pin2, LOW);  
  
}
```

Explanation of the Code

1. **Pin Setup:** We define pins for two motors connected to an L298N motor driver.
2. **Loop Function:** The loop function runs continuously, moving the robot in the desired pattern.
3. **Movement Functions:** There are four functions (`moveForward`, `moveBackward`, `turnRight`, and `turnLeft`) that set the motor direction pins to control the motors accordingly.
4. **Delay Times:** Each movement lasts 30 seconds (30,000 ms) before moving to the next command in the sequence.

UNIT 02 [Intermediate]

Chapter 04: Design Domain.

LESSON 01: Protect Your Components.

Time frame: 45 mins

Stepping Stone

You should know ...	You should be able to...
Basic navigation of the Tinkercad interface (e.g., view controls, toolbar functions)	Planning and designing a functional garbage box with appropriate dimensions
Understanding the concept of 3D space and axes (X, Y, Z)	Considering the size and placement of drainage holes for effective water flow
Placing and manipulating basic shapes (move, rotate, scale)	

Story

Apply the same strategies used in Part 1 of the story.

Think

1. How can we protect our electronic components from being damaged by water?

- Put the parts in a waterproof case or box.
- Cover the parts with a special waterproof coating.
- Place the electronics on a raised platform to keep them away from water.
- Use rubber or plastic seals to stop water from getting inside.
- Wrap the components with plastic or waterproof material to protect them.

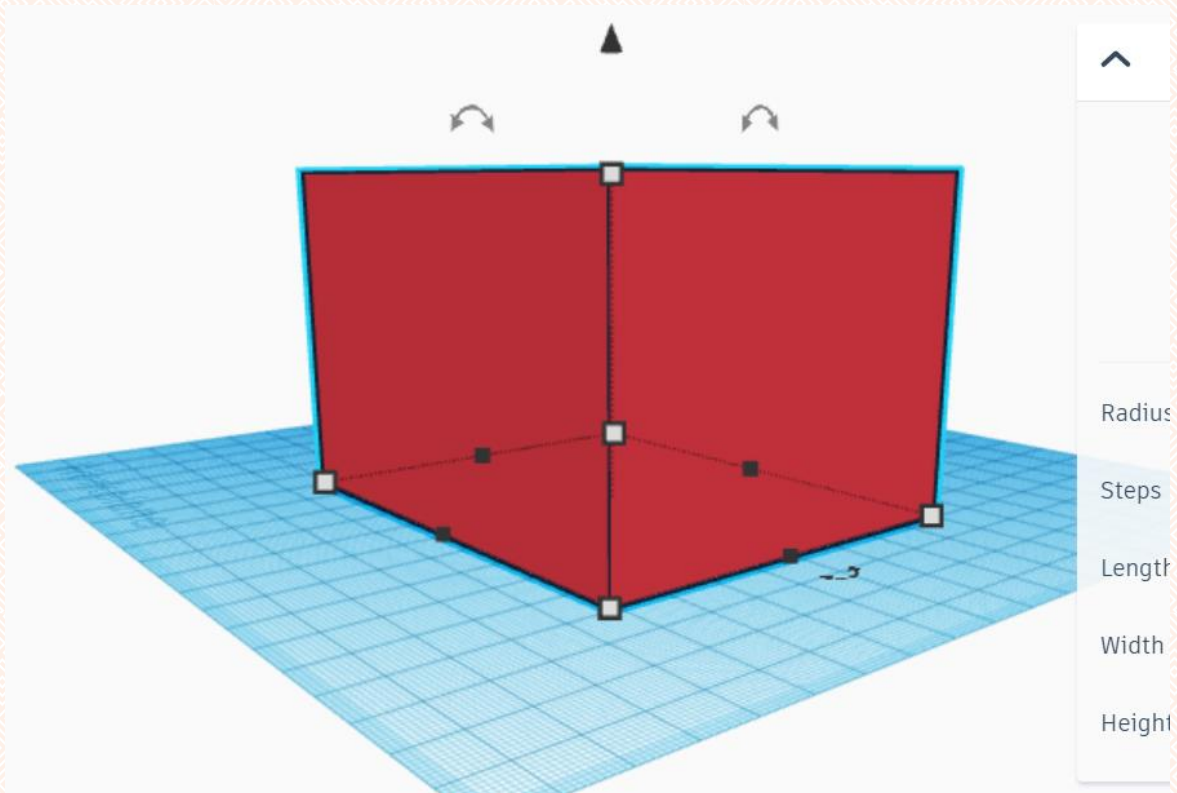
2. Where can we collect our garbage?

- Put it in the garbage bins or recycling bins.
- Use special collection areas set up by the school or classroom.
- Have a compost bin for food scraps or plant waste.
- Set aside a small area for collecting trash during the project.
- Use labeled bins for different types of waste, like plastic, paper, or food waste, to help with recycling.

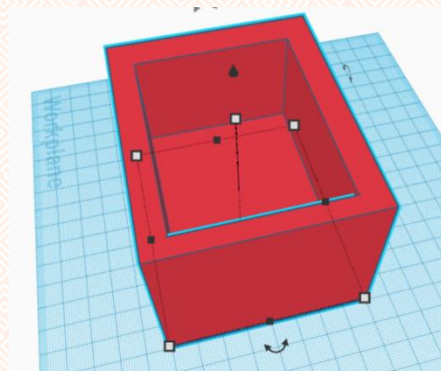
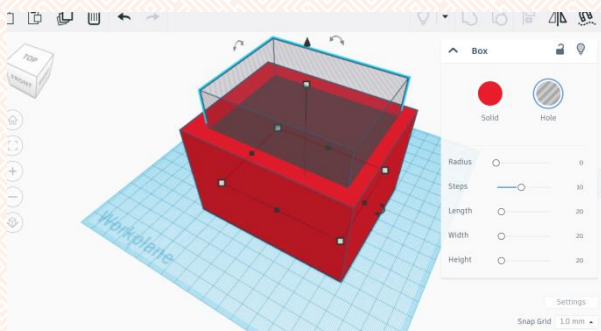
Explore

Do it...

1- Make a box that fits the dimensions of your floater.



2- Add a hole box inside it and group them together.



Assessment Exercise

Practice

Leave this part to the student's imagination.

EDP

Return back to the EDP paper and add the materials they are going to use to the improve part.



Show Case

Step 4: Box for the rubbish:

1. Cut Foam Pieces:

- Cut two rectangular pieces of foam, each measuring 3.5cm in length and 15cm in width.
- Cut one rectangular piece of foam measuring 3.5cm in length and 18cm in width.
- Cut one rectangular piece of foam measuring 10cm in length and 18cm in width.

2. Assemble the Rubbish Box:

- Use the glue gun or other adhesive to connect the foam pieces together to form a box shape. Refer to the provided image for guidance.



Attach to Sunboard:

- Once the box is dry, put it to the desired location on the sunboard.



UNIT 03 [Advanced]

Chapter 01: Programing.

LESSON 01: Design Your App.

Time frame: 45 mins

Stepping Stone

You should know ...	You should be able to...
Identify App inventor interface and how to make design for your application	How to control the floater with mobile
How to make simple code using app inventor	Understand the function of Bluetooth client in MIT App inventor

Story

Apply the same strategies used in Part 1 of the story.

Think

How can we operate or control the floater remotely by using the mobile phone?

Using Bluetooth:

- Connect a Bluetooth module to the floater's circuit.
- Use a mobile app that can send signals to the Bluetooth module to control the floater's movements.

Explore

DO it...

Now write answer in this part leave it to the student's imagination.

Watch it...

You may apply the same strategies previously used in this section.

Assessment Exercise

Focus

- **How does the application communicate with the floater?**

A. Bluetooth

The application typically communicates with the floater using Bluetooth, which allows wireless control at a close range.

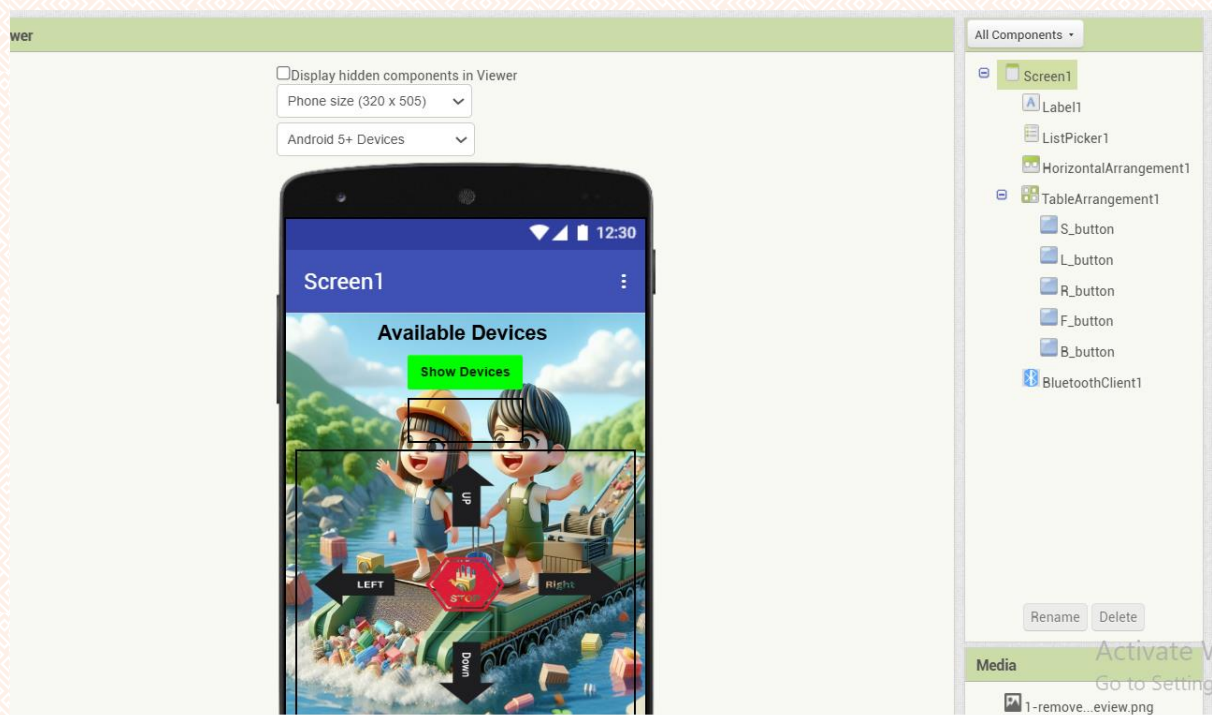
- **How do users interact with the floater through the application?**

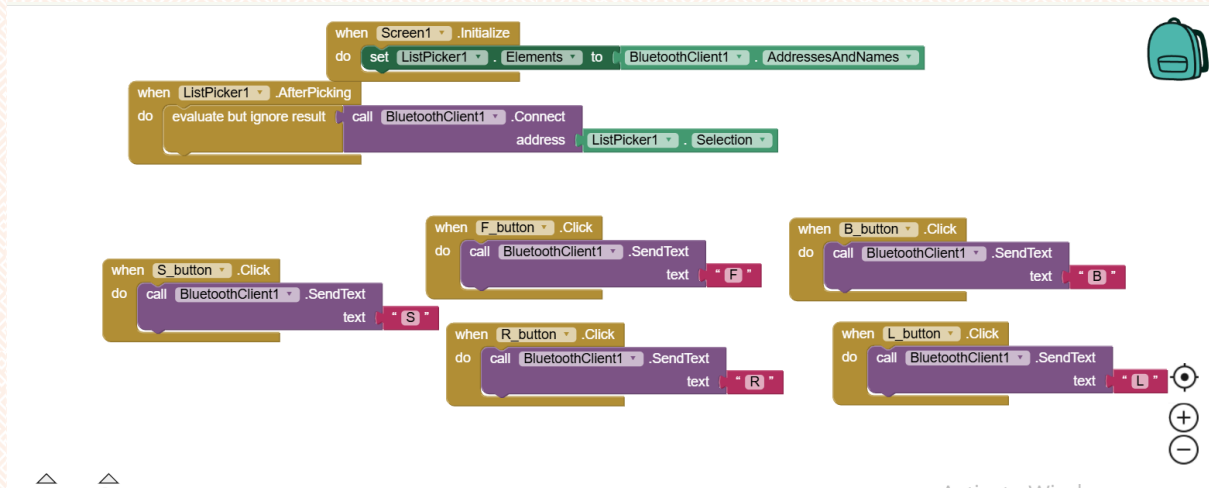
A. Touch gestures on the screen

Users interact with the floater by using touch gestures on the screen, such as pressing buttons or sliding controls within the app.

Practice

<https://drive.google.com/file/d/1CRC5U5jbVLQLj-Yy3itEWuXLCOKoLejd/view?usp=sharing>





EDP

Return back to the EDP paper and add the materials they are going to use to the improve part.



UNIT 03 [Advanced]

Chapter 02: Programing.

LESSON 01: Bluetooth connected.

Time frame: 45 mins

Stepping Stone

You should know ...	You should be able to...
Understanding of basic electronic components	Identifying the correct pins for connecting the Bluetooth module to the Arduino
Writing simple Arduino sketches (code)	Writing code to initialize and communicate with the Bluetooth module
Basic knowledge of digital and analog pins	Sending and receiving data via Bluetooth
Basic understanding of what Bluetooth is and how it works	

Story

Apply the same strategies used in Part 1 of the story.

Think

What innovative possibilities can be unlocked by using Bluetooth module in Arduino projects, and how can this transform everyday objects into smart devices?

- **Control Devices from a Distance:**
 - With Bluetooth, we can turn things on or off from our phones, like lights or fans. This makes it easier to control things around us without getting up!
- **See Information Right on Your Phone:**
 - We can use Bluetooth to send information (like the temperature or how bright it is) from our project to our phone. This helps us check things without looking directly at the project.
- **Make Things Adjust by Themselves:**
 - We can make everyday items, like lights or a fan, change automatically based on what we like, so they turn on or off by themselves!
- **Control Robots Wirelessly:**

- With Bluetooth, we can control robots or toy cars from our phones, making them move in different directions and do fun things.
- **Connect with Other Bluetooth Devices:**
 - Bluetooth can let our projects talk to other devices like speakers or headphones, so we can play sounds or music hands-free.
- **Help Keep Our Homes Safe:**
 - We can make a Bluetooth security system that sends an alert to our phone if something unusual happens, making our home safer.
- **Create Fun Fitness Tools:**
 - We can make devices that track our steps or heart rate and send the info to our phones. This way, we can learn more about how our body moves!
- **Make Learning Fun with Games:**
 - Bluetooth can let us make fun learning games that we control from a phone or tablet, like quizzes or flashcards, making learning more exciting.

Explore

Do it...

Simple Code to Communicate with Bluetooth Module

This code will send a message over Bluetooth when the Arduino receives any input from the connected device, such as a mobile app.

```
#include <SoftwareSerial.h>
```

```
// Define Bluetooth RX and TX pins
```

```
SoftwareSerial Bluetooth(10, 11); // RX = Pin 10, TX = Pin 11
```

```
void setup() {
```

```
  // Begin Serial communication with the computer
```

```
  Serial.begin(9600);
```

```
// Begin Bluetooth communication at 9600 baud rate

Bluetooth.begin(9600);

Serial.println("Bluetooth is ready to receive messages!");
}

void loop() {

  // If data is received from Bluetooth
  if (Bluetooth.available()) {
    char receivedChar = Bluetooth.read();
    Serial.print("Received: ");
    Serial.println(receivedChar);

    // Send a response back to Bluetooth
    Bluetooth.println("Hello from Arduino!");
  }

  // If data is received from Serial Monitor (for testing)
  if (Serial.available()) {
    char message = Serial.read();
    Bluetooth.print(message);
  }
}
```


Explanation of the Code:

- `SoftwareSerial Bluetooth(10, 11);` — Defines the RX and TX pins for the Bluetooth communication.
- `Bluetooth.begin(9600);` — Starts Bluetooth communication.
- `Bluetooth.read();` — Reads incoming data from the Bluetooth module.
- `Bluetooth.println("Hello from Arduino!");` — Sends a message back to the connected device.

How it Works:

- When a message is sent from a Bluetooth device (like a phone), the Arduino reads it and prints it to the Serial Monitor.
- Then, it sends a response message back to the Bluetooth device.

Watch it...

You may apply the same strategies previously used in this section.

Assessment Exercise

Focus

- **What is the primary purpose of a Bluetooth module in an Arduino project?**

Answer: B) To enable wireless communication

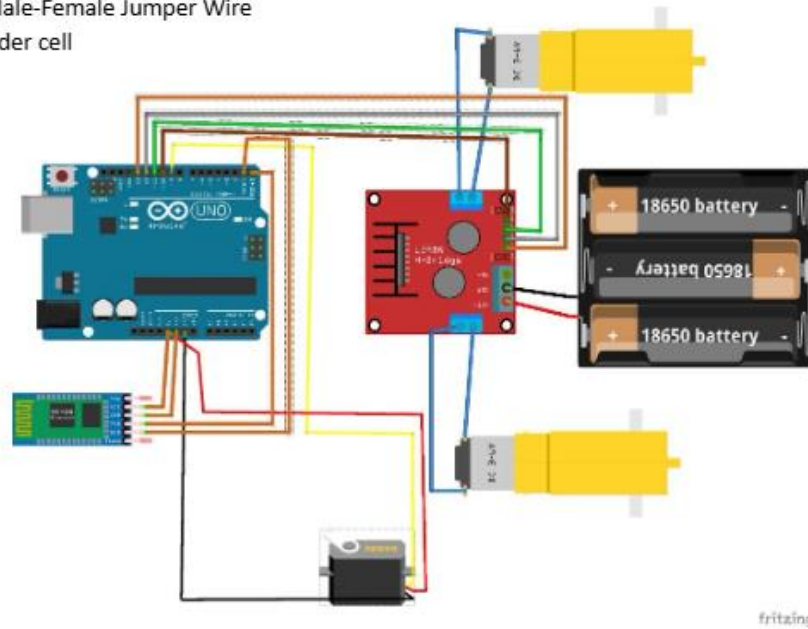
- **Which of the following is a common Bluetooth module used with Arduino?**

Answer: A) HC-05

Practice

Materials used:

- Arduino uno
- Servo motor
- Motor driver
- Bluetooth module
- 2 dc motor yellow
- 3 li battery
- 9 Male-Male Jumper Wire
- 3 Male-Female Jumper Wire
- Holder cell



Code

```
#include <Servo.h>

// Define servo pin
const int servoPin = 9;

Servo steeringServo;

// Variables to store received command
char t;

void setup() {
    // Set motor pins as outputs
    pinMode(13,OUTPUT); //left motors forward
    pinMode(12,OUTPUT); //left motors reverse
    pinMode(11,OUTPUT); //right motors forward
    pinMode(10,OUTPUT); //right motors reverse
```



```

// Start serial communication for Bluetooth module
Serial.begin(9600);
}

void loop() {
    // Check if data is available on the serial port
    if (Serial.available() > 0) {
        t = Serial.read(); // Read the incoming data
        Serial.println(t);
    }

    // Handle the received command
    if(t == 'F'){        //move forward(all motors rotate in forward direction)
        digitalWrite(13,HIGH);
        digitalWrite(11,HIGH);
    }

    if(t == 'B'){        //move reverse (all motors rotate in reverse direction)
        digitalWrite(12,HIGH);
        digitalWrite(10,HIGH);
    }

    const int initialServoPosition = 90;

    if(t == 'L'){        //turn right (left side motors rotate in forward direction, right side motors doesn't rotate)
        steeringServo.write(180); // Turn servo left
    }

    if(t == 'R'){        //turn left (right side motors rotate in forward direction, left side motors doesn't rotate)
        steeringServo.write(0); // Turn servo right
    }

    if(t == 'S'){        //STOP (all motors stop)
        digitalWrite(13,LOW);
        digitalWrite(12,LOW);
        digitalWrite(11,LOW);
    }
}

```

```
digitalWrite(10,LOW);
```

```
}
```

```
}
```

Show Case

Test the floater in water.

Create your Story

In this part the teacher should print photos through the whole project and glues it in the student's books.